

An
Internship Report
on
“Food Trends Understanding Customer
Preferences in Food and Beverages”

BY

Ms. Pooja More

Under the Guidance

of

Mr. M. N. Bhandare



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
Mahatma Gandhi Mission's College of Engineering, Nanded (M.S.)

Academic Year 2025–26

An Internship Report on
“Food Trends Understanding Customer
Preferences in Food and Beverages”

Submitted to
DR.BABASAHEB AMBEDKAR TECHNOLOGICAL
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in partial fulfilment of the requirement for the degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING

By
Ms. Pooja More

Under the Guidance
of
Mr. M. N. Bhandare

(Department of Computer Science & Engineering)



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING,
NANDED (M.S.)

Academic Year 2025-26

Certificate



This is to certify that the project entitled

“Food Trends Understanding Customer Preferences in Food and Beverages”

*being submitted by **Ms. Pooja More** to the Dr. Babasaheb Ambedkar Technological University, Lonere, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by her under my supervision and guidance. The matter contained in this report has not been submitted to any other university or institute for the award of any degree.*

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INTERNSHIP OFFER LETTER & CERTIFICATE



Dear Learner,

With reference to the **Infosys Springboard Internship 6.0**, you should have now received your project title and meeting details. Please attend all scheduled meetings as per the communicated plan. Ensure that you attend every session and complete your assigned tasks on time. **If you have not received any communication, kindly reach out to your mentor.**

Do not miss this valuable opportunity to complete your virtual internship.

Attendance and individual contribution to project is necessary to qualify for internship completion certificate. If you need to take leave for academic exams or placement drives, obtain written approval via email beforehand. If you haven't received the meeting link or timing, kindly reach out to your mentor.

Project title: FoodTrends Understanding Customer Preferences in FB

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Got a query? Write to springboard-support@infosys.com

Regards,
Team Infosys Springboard

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Dear Intern,

Congratulations! You have successfully completed Infosys Springboard Internship 6.0! We're proud of your achievement and the dedication you've shown throughout the internship.

Your certificate will be sent to your registered email address shortly. Keep up the great work and continue striving for excellence!

Announce your achievement and your internship experience on social media. Tag us @InfosysSpringboard when you do.

Note: Your project is your asset. Refrain from sharing your artifacts in online forums/domains to safeguard your work's ownership.

The name on the certificate will be as per the profile name registered on the Infosys Springboard platform.

Regards,
Team Infosys Springboard

Keep your learning journey going. Scan the QR code to download the Infosys Springboard App and access your courses on the go!



App Store



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ACKNOWLEDGMENT

I would like to express my deepest gratitude to my project guide, **Mr. M. N. Bhandare**, for his invaluable support, guidance, and encouragement throughout this project. His profound knowledge and expertise have been instrumental in the successful completion of this work. His patience and willingness to assist me at every step have greatly enriched my learning experience. His constructive feedback and insightful suggestions have not only helped me overcome challenges but also motivated me to strive for excellence.

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I would also like to extend my sincere gratitude to Infosys Springboard for providing me with the opportunity to undertake the Infosys Springboard Virtual Internship 6.0 and gain valuable industry experience. I am deeply thankful to my internship mentor, **Mrs. Kalaiyarasi Gomathi**, whose expert guidance, constant support, and practical insights greatly contributed to my professional growth and helped me bridge the gap between academic learning and real-world application. The experience and knowledge gained during this internship have been an integral part of this project's development.

Last but not least, we are also thankful to all those who helped, directly or indirectly, develop this project and complete it successfully.

With Deep Reverence,

Ms. Pooja More
[B.Tech CSE-B]

ABSTRACT

The food and beverage industry in India is really growing fast. It is changing all the time because people want to eat things. The internet is making it easy for people to order food from home. More and more people are moving to cities. The food and beverage industry has a lot of competition between restaurants and the companies that deliver food. The food and beverage industry is one of the growing industries, in India. Understanding what customers prefer, how they engage, and what drives their satisfaction has become a critical business need for restaurant operators, food-tech companies, and market strategists alike. This project — "Food Trends: Understanding Customer Preferences in Food & Beverages" — addresses this need by presenting a comprehensive, multi-page interactive analytics dashboard built using Microsoft Power BI, designed to decode customer behavior, sentiment patterns, product preferences, and market performance across major Indian cities. The project is powered by a unified dataset Fact Orders, capturing restaurant and order-level information including cuisines, pricing, delivery time, ratings, sentiment, revenue, order frequency, and user engagement — enabling cross-dimensional analysis across customer, product, operations, and geography.

The dashboard spans seven analytical pages. The Executive Overview presents critical KPIs — Total Restaurants, Total Orders, Total Cities, Average Rating, Average Price, and Average Delivery Time — for an instant market health check. Sentiment Analysis classifies feedback into Positive, Neutral, and Negative categories to identify customer trust levels. Product Insight & Preference Analysis reveals ordering patterns through Top 10 Cuisines, Cuisine Market Share, and Average Order Value by Price Segment. Customer Segmentation profiles customers by Order Frequency, Engagement, and Satisfaction for targeted retention strategies. Strategy & Innovation uses Power BI's AI-powered Key Influencers visual to identify variables driving customer satisfaction. Operations & Delivery Performance links Fast Delivery Percentage and Delivery Performance Score to satisfaction outcomes. Finally, Geographic & Market Analysis delivers a City vs. Cuisine Heatmap and Market Rankings to support expansion decisions.

In conclusion, this dashboard delivers a 360° view of India's food and beverage landscape, empowering stakeholders with data-driven insights to optimize menus, pricing, delivery, and market expansion for sustained business growth.

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INTRODUCTION

Internships play a vital role in bridging the gap between theoretical academic learning and real-world industry exposure. As part of my final semester of the B.Tech in Computer Science and Engineering program, I had the opportunity to undertake a virtual internship under the Infosys Springboard Virtual Internship 6.0 program. This experience allowed me to engage with industry-standard tools, practices, and methodologies in the field of Data Visualization, all within a structured and professionally guided learning environment. The internship provided a platform to enhance my technical skills and analytical capabilities while contributing to a data-driven project designed to uncover meaningful insights from real-world food and beverage industry data. Infosys Springboard is a prominent digital learning initiative by Infosys, one of India's leading multinational technology companies. The Springboard platform is recognized for its innovative approach to skill development, offering learners access to curated courses, industry mentorship, hands-on projects, and virtual internship experiences across various domains including data science, artificial intelligence, cloud computing, and data visualization. The program is really good at helping people learn and get better at what they do. It is a place for people who want to work with data to get some real experience. When I was an intern I got to work on a project that was about data. I had mentors who work in the field. They helped me. I was in charge of making a dashboard that people could use to look at data in a fun way.

Before I started working on the project I went through a training program. This program taught me the basics and some complicated things about making data look good business tools and data analysis. I learned a lot about Microsoft Power BI. This includes making models with data using DAX measures designing reports and telling stories, with data in a way that's interactive and easy to understand. The data visualization and business intelligence tools were really interesting to learn about. I feel like I got a lot out of the program. It helped me become a better data professional. I also developed skills in data cleaning, data transformation, and building multi-page dashboards that communicate complex insights in a clear and actionable manner. This foundation was crucial for my subsequent contributions to the internship project. During the internship, I applied the skills gained from the training phase to develop a comprehensive Power BI dashboard titled "Food Trends: Understanding Customer Preferences in Food & Beverages." I was responsible for designing and building an interactive multi-page report that analyzes customer behavior, sentiment patterns, product preferences, delivery performance, and geographic market trends across major Indian cities. The dashboard was set up to give restaurant operators and food tech companies and people who make business

plans the information they need to make decisions. It does this by using pictures and numbers to tell a story. The dashboard is supposed to help restaurant operators and food tech companies and business strategists by making it easy for them to understand what is going on.

1.1 Internship Overview

An internship forms one of the most essential aspects of an engineering student's learning journey, acting as a vital bridge between classroom learning and practical industry experience. I think internships are really great for students because they get to see what it is like in the world. Students get to use tools and work with other people, which is something they might not get to do in school. They also get to see what it is like to work in a setting, which is very different from school. The things I learned during my internship I could not have learned in school. That makes it a very important part of my education. As part of my semester of B.Tech in Computer Science and Engineering I did the Infosys Springboard Virtual Internship 6.0 which was about Data Visualization.

Infosys Springboard is a known platform where students can learn and get work experience from home. The platform provides students with work experience through projects and guidance from mentors. I was really excited to work with Data Visualization because it is used to make decisions. It was also very interesting and fun to learn about. I got to contribute to the project and learn about Data Visualization at the same time. The internship was really practical. I worked with data and created dashboards that helped fix issues in the food and beverage industry. I started with tasks and then moved on to tougher ones. The internship was set up to help me learn step by step. I made interactive dashboards to solve problems, in the food and beverage industry. It was an experience working with real data.

The internship was set up so that I started with tasks and then moved on to more complex ones. It started with training, on Data Visualization and Power BI tools. Then I got to work on a real project. The Infosys Springboard program was very helpful because I got to learn from modules, project reviews and mentorship sessions. This helped me understand how Data Visualization projects are planned, made and finished in work settings.

1.2 Company Profile

Infosys Limited is a big and well known technology company in India. The company is based in Bengaluru, Karnataka. Infosys Limited was started in the year 1981. It has become a big name in technology services, consulting and digital transformation. The company works with clients in than 50 countries. Infosys is known for trying things doing business in a fair way and helping its employees get better at what they do. The

company is always looking for ways to use technology to help businesses deal with tough problems and grow in a way that is good for the long term. Infosys is really good at using technology to help businesses solve challenges and achieve sustainable growth, which is what Infosys is all, about using technology to help businesses grow.



Fig 1.1: Company Logo

Infosys Springboard is a program that Infosys started to help people learn new skills and get experience working on projects. This program is like a bridge between what people learn in school and what they need to know to work in a company. The Infosys Springboard platform gives people a chance to take a lot of courses and get certifications in many areas, such as data science and artificial intelligence and machine learning. They also get to work on projects that're like real jobs in areas, like cloud computing and cybersecurity and data visualization. Infosys Springboard is really helping people get the skills they need to work in these fields.

The Infosys Springboard Virtual Internship 6.0 is specifically designed to give students hands-on exposure to industry-relevant projects under the mentorship of experienced professionals, equipping them with the skills and confidence needed to succeed in the modern technology workforce. During my time with the Infosys Springboard program, I worked within the Data Visualization domain, where I was assigned to develop a comprehensive business intelligence dashboard using Microsoft Power BI. The program's emphasis on practical learning, structured milestones, and real-world datasets provided me with a rich and stimulating experience. Infosys Springboard's forward-looking vision and culture of excellence made my internship experience extremely rewarding and educational.

1.3 Project Objectives

The objective of this project is to design, develop, and implement a comprehensive and interactive business intelligence dashboard using Microsoft Power BI to analyze and interpret customer preferences, operational efficiency, and market trends in the food and beverage industry. In today's data-driven environment, organizations generate vast amounts of transactional and behavioral data; however, extracting meaningful insights from this data requires advanced analytical tools and structured visualization techniques. This project aims to bridge that gap by transforming raw, unstructured data into a well-organized and insightful visual format that supports effective decision-making. The dashboard developed in this project focuses on multiple analytical dimen-

sions such as customer sentiment, product demand, pricing patterns, delivery performance, and geographic distribution. By leveraging Power BI's capabilities, including data modeling, DAX-based calculations, and interactive visualizations, the project provides a unified platform where complex datasets can be explored in a simple and intuitive manner. The objective is not only to present data but to tell a coherent data story that highlights key trends, patterns, and relationships within the dataset. Furthermore, the project aims to demonstrate the practical application of data visualization principles and business intelligence tools in solving real-world problems. It emphasizes the importance of data preprocessing, accurate metric calculation, and effective dashboard design in delivering reliable insights. The inclusion of interactive features such as slicers, filters, and drill-through functionality enables users to customize their analysis and gain deeper insights according to their specific requirements. Overall, the project seeks to enhance analytical understanding, improve decision-making capabilities, and provide a scalable solution that can be extended for future analytical needs in the food and beverage sector.

In addition to analytical and visualization objectives, the project also focuses on developing a systematic and structured approach to handling real-world datasets and delivering insights in a professional format. It highlights the importance of organizing data workflows, from initial data cleaning and transformation to final dashboard presentation, ensuring accuracy and consistency at every stage. The project further aims to build the ability to interpret data in a business context by connecting analytical. Our research has some really useful findings that can help make customers happier, make our operations run more smoothly, and overall make things better strategic planning. By integrating technical skills with analytical thinking and visualization techniques, the project serves as a comprehensive learning experience that prepares for real-world applications in the field of data analytics and business intelligence.

1.4 Problem Statement

The food and beverage industry operates in a highly dynamic and competitive environment where customer preferences, ordering patterns, and service expectations are constantly evolving. With the rapid growth of digital food delivery platforms and online ordering systems, a massive volume of data is generated on a daily basis, including information related to customer behavior, restaurant performance, pricing structures, delivery efficiency, and geographic demand patterns. While this data holds significant potential for improving business performance and enhancing customer satisfaction, it often remains underutilized due to the lack of effective data analysis and visualization mechanisms. Organizations frequently rely on traditional data analysis methods such as spreadsheets and static reports, which are not capable of handling large-scale datasets

or uncovering complex relationships within the data. One of the primary challenges faced by businesses in this domain is the difficulty in extracting meaningful insights from raw and unstructured data.

The data is often scattered across multiple dimensions such as customer ratings, delivery time, pricing levels, cuisine preferences, and location-based variations, making it difficult to analyze these factors collectively. As a result, decision-makers are unable to gain a comprehensive understanding of the key drivers influencing customer satisfaction and business performance. For example, while a business may observe a decline in customer ratings, it may not be able to identify whether the issue is related to delivery delays, pricing strategies, or product quality without a unified analytical approach. Another significant problem is the absence of an integrated platform that allows for multidimensional analysis. Businesses need to simultaneously evaluate various aspects such as sentiment analysis, product demand, To make smart choices, you need to look at how well things are running, how different areas are doing, and how they're performing overall. However, without a centralized system that combines these analytical perspectives, it becomes challenging to identify trends, correlations, and patterns that can guide strategic planning. This lack of integration often results in fragmented insights, delayed decision-making, and missed opportunities for optimization and growth.

Furthermore, the lack of interactivity in conventional reporting systems limits the ability of users to explore data according to their specific requirements. Static reports do not provide flexibility for filtering, drilling down into details, or comparing different segments of data, which restricts the depth of analysis. Decision-makers require interactive tools that allow them to dynamically explore data, identify anomalies, and derive insights in real time. Without such capabilities, organizations are unable to respond quickly to changing market conditions or customer needs. This project addresses these challenges by developing an interactive and user-friendly dash- board using Microsoft Power BI that integrates multiple data dimensions into a single analytical platform. The solution involves cleaning and transforming raw data, creating meaningful metrics using DAX, and designing intuitive visualizations that clearly communicate insights. The dashboard enables users to analyze customer preferences, evaluate product performance, assess delivery efficiency, and explore geographic trends through interactive features such as slicers, filters, and drill-through options. By providing a structured and visually rich representation of data, the project bridges the gap between raw information and actionable business intelligence, thereby supporting better decision-making and improving overall operational effectiveness.

1.5 Scope of the Project

The scope of this project extends across the complete lifecycle of designing and

implementing a business intelligence and data visualization solution using Microsoft Power BI, with a strong focus on transforming raw and unstructured data into meaningful, actionable insights. The project is centered around the analysis of a real-world food and beverage dataset, which includes detailed information related to customer orders, restaurant attributes, pricing, customer ratings, delivery time, and geographic distribution. The primary aim within this scope is to systematically process, analyze, and visualize this data in a structured and interactive manner that supports effective decision-making. The initial scope of the project involves comprehensive data preparation and preprocessing using Power Query. This includes identifying and resolving data quality issues such as missing values, duplicate records, and inconsistent formatting. Data transformation steps are applied to standardize the dataset, ensuring that all variables are correctly formatted and ready for analysis. In addition, new derived attributes are created, such as price segments, rating categories, delivery performance classifications, and sentiment labels, which enhance the analytical depth of the dataset and enable multi-dimensional analysis. Getting good results from analysis depends on doing some important steps first. These steps help make sure the information we get in the end is reliable and accurate. Another key component within the scope is the development of a robust data model.

This project is all about setting up a dataset so it's easy to search and interact with. We want to make sure the data is organized in a way that makes it simple to get the information we need, quickly and smoothly. between different data attributes. This includes defining relationships between data elements, organizing data hierarchies, and ensuring that the model supports dynamic filtering and aggregation. The use of Data Analysis Expressions (DAX) further extends the analytical capabilities of the project by enabling the creation of calculated columns and measures. These calculations are used to derive key performance indicators such as total number of orders, average customer rating, average pricing, delivery efficiency metrics, and customer satisfaction distribution, which form the core of the dashboard analysis. A major part of the project scope involves the design and development of a multi-page interactive dashboard. Each page of the dashboard is dedicated to a specific analytical perspective, including executive overview, customer sentiment analysis, product and cuisine insights, customer segmentation, strategic insights using advanced visuals, delivery performance evaluation, and geographic analysis. The dashboard is designed following data visualization best practices, ensuring clarity, consistency, and effective communication of insights. Interactive features such as slicers, filters, drill-down capabilities, and cross-filtering are incorporated to allow users to explore the data dynamically and gain deeper insights based on their specific analytical needs.

Furthermore, the project scope includes the application of data storytelling principles to ensure that the good dashboard doesn't just show you a bunch of numbers and

charts, it actually tells a story with the data, making it easy to understand and follow. The arrangement of dashboard pages and visuals is structured in a way that guides the user from high-level summary insights to more detailed analytical perspectives, enabling a comprehensive understanding of the dataset. The project also demonstrates the integration of analytical thinking and visualization techniques to uncover patterns, correlations, and trends that may not be immediately visible in raw data form. However, it is important to note that the scope of this project is limited to descriptive and exploratory data analysis using historical data. It doesn't cover things like predicting what might happen in the future, using special computer programs to make predictions, or combining data from different sources right away. The project doesn't require us to set it up in a real-world setting or connect it to other systems external systems.

1.6 Organization of the Report

This report is structured into five chapters:

Chapter 1: Introduction – Provides an overview of the Infosys Springboard Virtual Internship 6.0 in Data Visualization, outlining the objectives, significance of data analytics, and the training process followed during the internship. It also highlights the tools used and introduces the Food Trends Power BI dashboard project along with the key learning outcomes achieved.

Chapter 2: Company Background – Describes Infosys Limited as an organization, covering its history, global presence, vision, and mission. It also explains the Infosys Springboard platform and outlines the structure, learning approach, and objectives of the Virtual Internship 6.0 program, emphasizing its role in industry-oriented skill development.

Chapter 3: Technology Stack – Explains the technologies used, including Microsoft Power BI, DAX, Power Query, and data modeling techniques. It further describes their role in data cleaning, transformation, analysis, and visualization, along with their practical application in developing the Food Trends dashboard.

Chapter 4: Technical Training and Assignment – Summarizes the training modules such as Power BI fundamentals, DAX calculations, Power Query transformations, dashboard design principles, and data storytelling techniques. It also includes the practical assignments completed during the internship, which helped in building hands-on experience.

Chapter 5: Internship Project Contribution – Describes the end-to-end development of the Food Trends dashboard, including data preprocessing, data modeling, and KPI creation. It further explains sentiment analysis, customer segmentation, delivery performance evaluation, geographic analysis, key insights derived, and challenges faced during the project implementation.

COMPANY BACKGROUND

Infosys Limited is one of India's most celebrated and globally recognized multinational technology companies, with a legacy spanning over four decades. Since its founding in 1981 by N. R. Narayana Murthy and six co-founders — Nandan Nilekani, S. Gopalakrishnan, S. D. Shibulal, K. Dinesh, N. S. Raghavan, and Ashok Arora — with an initial capital of just \$250, Infosys has grown into a global powerhouse in technology services, consulting, and digital transformation. Headquartered in Bengaluru, Karnataka, India, the company today employs over 300,000 professionals across more than 50 countries, serving clients in industries ranging from banking and financial services to retail, healthcare, manufacturing, energy, and telecommunications. Infosys is listed on both the Bombay Stock Exchange and the National Stock Exchange in India, as well as on the NASDAQ in the United States, reflecting its stature as a truly global enterprise. The company's journey from a small software firm in Pune to a Fortune 500 company is a testament to its unwavering commitment to innovation, ethical business practices, and human capital development. Infosys has consistently been ranked among the world's most admired companies and has received numerous awards and recognitions for its contributions to the technology industry, corporate governance, sustainability, and employee welfare. Its client roster includes some of the world's largest corporations, and its solutions have helped businesses across the globe navigate complex digital transformations and achieve sustainable competitive advantage. The company's revenues have grown from a modest beginning to billions of dollars annually, establishing Infosys as a symbol of India's emergence as a global technology powerhouse and a model for how Indian enterprises can compete and excel on the world stage.

Over the years, Infosys has expanded its global footprint significantly, establishing development centers, offices, and innovation hubs across North America, Europe, Asia Pacific, the Middle East, and Africa. This global presence allows Infosys to serve its clients with local expertise and cultural understanding while leveraging the depth of its global talent pool and technological capabilities. The company has also made strategic acquisitions of leading technology firms across the world to strengthen its capabilities in areas such as artificial intelligence, cloud computing, digital experience, and industry-specific solutions, ensuring that it remains competitive and relevant in an ever-evolving technology landscape.

2.1 About Infosys Limited

Infosys Limited operates as a global leader in next-generation digital services and consulting, helping clients in more than 56 countries navigate their digital transforma-

tion journeys. The company offers a comprehensive portfolio of services including application development and maintenance, business process management, cloud services, cybersecurity, data analytics, artificial intelligence, enterprise resource planning, and engineering services. Infosys combines deep domain expertise with cutting-edge technology capabilities to deliver solutions that are innovative, scalable, and aligned with the strategic objectives of its clients. The company is organized into multiple business segments, each specializing in a distinct area of technology and industry expertise. The Digital segment focuses on helping clients harness the power of emerging technologies such as artificial intelligence, machine learning, blockchain, and the Internet of Things to create new business models and digital experiences. The Cloud segment assists organizations in migrating their infrastructure and applications to the cloud, optimizing cloud operations, and building cloud-native solutions that are scalable, resilient, and cost-efficient. The Core segment delivers traditional IT services such as application management, infrastructure management, and business process outsourcing, which remain a key component of Infosys's revenue stream.

Infosys invests heavily in research and development through its Infosys Labs division, which focuses on emerging technologies and their practical applications in solving real-world business problems. Infosys Labs collaborates with universities, research institutions, and technology partners across the globe to stay at the cutting edge of innovation and to develop intellectual property that creates differentiated value for clients. The company's workforce is one of its greatest strengths, and it invests significantly in the continuous learning and development of its employees through programs such as Infosys Lex, the company's internal learning platform, which provides employees access to thousands of courses, certifications, and learning resources. Infosys also places a strong emphasis on corporate social responsibility through the Infosys Foundation, undertaking initiatives in education, healthcare, rural development, arts and culture, and disaster relief across India, while actively pursuing carbon neutrality and promoting diversity and inclusion across its global workforce.

2.2 Vision and Mission

The vision of Infosys is to be a globally respected corporation that provides best-in-class business solutions, leveraging technology, and delivering value to its clients, employees, and stakeholders. This vision shows what the company wants to be: not just a company that provides technology services, but a company that people trust. long-term partner that helps businesses transform, innovate, and grow in an increasingly digital and interconnected world. It embodies the belief that technology, when applied thoughtfully and responsibly, has the power to create. It offers really great benefits for people, companies, and the community at large. The mission of Infosys is to achieve

its objectives in an environment of fairness, honesty, and courtesy towards its clients, employees, vendors, and society at large. This mission statement encapsulates the core values that have guided Infosys since its inception integrity. We strive for respect, excellence, and a strong sense of responsibility to everyone involved. These values are not merely words on paper but are actively practiced and reinforced at every level of the organization, from the boardroom to the development floor. They define how Infosys engages with its clients, how it treats its employees, how it selects and works with its vendors, and how it contributes to the communities in which it operates.

Infosys is guided by a set of core values collectively known as C-LIFE, which stands for Client Value, Leadership by Example, Integrity and Transparency, Fairness, and Excellence. Client Value reflects the company's commitment to always putting the client's interests at the center of it makes sure that everything it does creates value that can be measured and lasts. Leadership by Example emphasizes the importance of leaders at all levels setting the standard for professional behavior, ethical conduct, and commitment to excellence. Integrity and Transparency underscore the company's belief that trust is the foundation of every successful relationship, and that honesty and openness are non-negotiable in all interactions. Fairness reflects Infosys's commitment to treating every individual — whether an employee, client, or vendor with equal respect, dignity, and opportunity.

Excellence represents the relentless pursuit of the highest standards of quality, innovation, and performance in everything the company undertakes. These values serve as the ethical and professional compass for every Infosys employee and define the culture. This has helped the company become a well-known and reliable name in the technology industry around the world. The emphasis on leadership by example and continuous excellence ensures that Infosys consistently delivers. We deliver results that go beyond what our clients expect, all while keeping our work at the highest level of professionalism. The C-LIFE values are the backbone of Infosys's work environment and also play a big role in how they design things and delivery of its products, services, and learning initiatives — including the Infosys Springboard platform ensuring that everything the company does is aligned with its commitment to creating value, fostering integrity, and pursuing excellence.

2.3 Infosys Springboard Platform

Infosys Springboard is a flagship digital skilling and lifelong learning platform launched by Infosys with the vision of empowering one million learners across India with industry-relevant skills and knowledge. The platform came about because of a big problem in India - not enough people have the digital skills they need. This is happening because technology is moving so fast advancement has created a significant mis-

match between the skills that educational institutions impart and the competencies that the modern technology industry demands. Infosys Springboard reflects the company's deep commitment to giving back to society by democratizing access to quality technology education and bridging the divide between academic learning and real-world industry requirements. The platform offers a rich and diverse catalog of learning content spanning a wide range of domains including data science, artificial intelligence, machine learning, cloud computing, cybersecurity, full-stack web development, data visualization, business analytics, digital marketing, and design thinking.

Learners on the Springboard platform have access to content in multiple formats including video lectures, interactive exercises, reading materials, quizzes, and hands-on projects, catering to different learning styles and preferences. Our content is carefully chosen and checked by experts in the field from Infosys and other companies we work with, making sure it's accurate and helpful. It needs to be accurate, up-to-date, and relevant to what's actually happening in the tech industry right now, so it can really prepare people for the workforce.

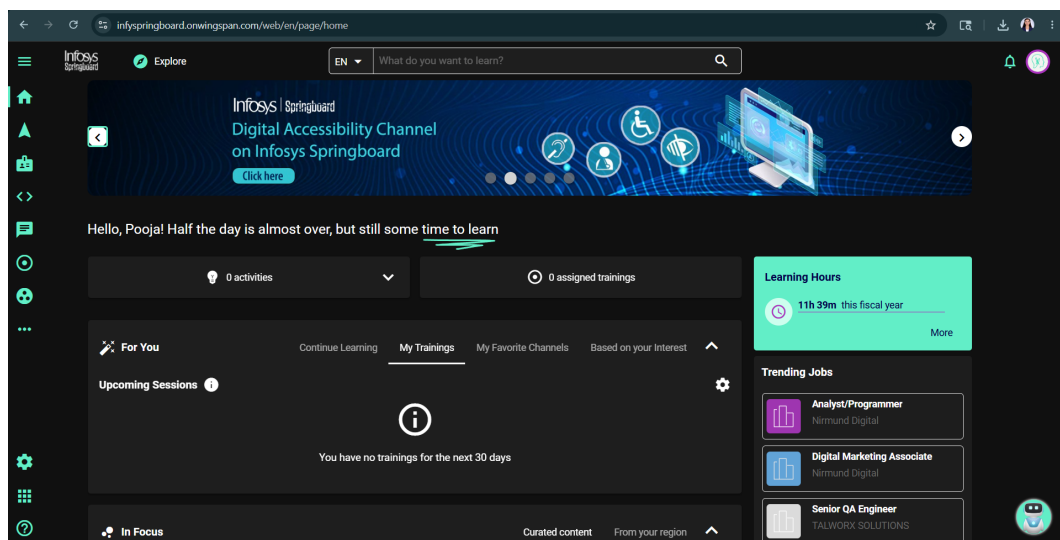


Fig 2.1: Infosys Springboard Platform

Infosys Springboard is available free of charge to students and learners across India, removing financial barriers that might otherwise prevent talented individuals from accessing quality technology education. The platform supports learners at different stages of their educational journey, from school students exploring technology for the first time to college students preparing for their first job and working professionals looking to upskill or reskill. One of the most distinctive features of Infosys Springboard is its emphasis on experiential learning, integrating real-world projects, industry case studies, simulation exercises, and virtual internship programs that allow learners to apply their knowledge in environments that closely simulate professional work settings. The platform also gives learners a way to get certifications that are recognized by industries, which proves they have the skills they need. This helps them stand out and have a real

advantage when looking for a job, making it easier to get hired. Additionally, Infosys Springboard partners with colleges and universities across India to integrate its learning content into formal curricula, extending its reach to millions of students and creating a multiplier effect on the broader education ecosystem.

2.4 Virtual Internship 6.0 Program Structure

The Infosys Springboard Virtual Internship 6.0 is a structured, project-based virtual internship program designed specifically for engineering and technology students across India. It is the sixth and most refined edition of the Virtual Internship series under the Springboard initiative, incorporating improvements based on feedback from previous cohorts, advances in technology, and changing industry needs. The program is organized across multiple domains including data visualization, data science, artificial intelligence, machine learning, full-stack development, cloud computing, and cybersecurity, with students assigned to a domain based on their academic background, interests, and career aspirations. Each domain is associated with a specific real-world project that reflects actual industry challenges and requires students to apply a combination of technical skills, analytical thinking, and problem-solving capabilities to produce a professional-grade deliverable. The Virtual Internship 6.0 program follows a well-defined structure divided into distinct phases. The program begins with a comprehensive onboarding phase during which students are introduced to the program's objectives, structure, expectations, evaluation criteria, and available learning resources, ensuring that all participants begin with a clear understanding of what is expected of them and what they will gain from the experience. Following onboarding, the program transitions into the structured training phase, where students complete a series of domain-specific learning modules covering both foundational and advanced concepts relevant to their area of focus. The training is designed to be progressive and scaffolded, beginning with core principles and gradually introducing more complex and applied topics as students build confidence and competence. For the Data Visualization domain, the training covered Microsoft Power BI fundamentals, data import and transformation using Power Query, data modeling and relationship management, DAX formula writing for calculated measures and KPIs, dashboard design best practices, interactive report development, and data storytelling techniques.

The project phase is the core and most substantial component of the Virtual Internship 6.0 program, during which students work independently on their assigned domain project, applying the skills and knowledge developed during training to produce a complete and professionally presented deliverable. In the Data Visualization domain, the assigned project was the development of a comprehensive interactive dashboard using Microsoft Power BI, analyzing a real-world food and beverage industry dataset and de-

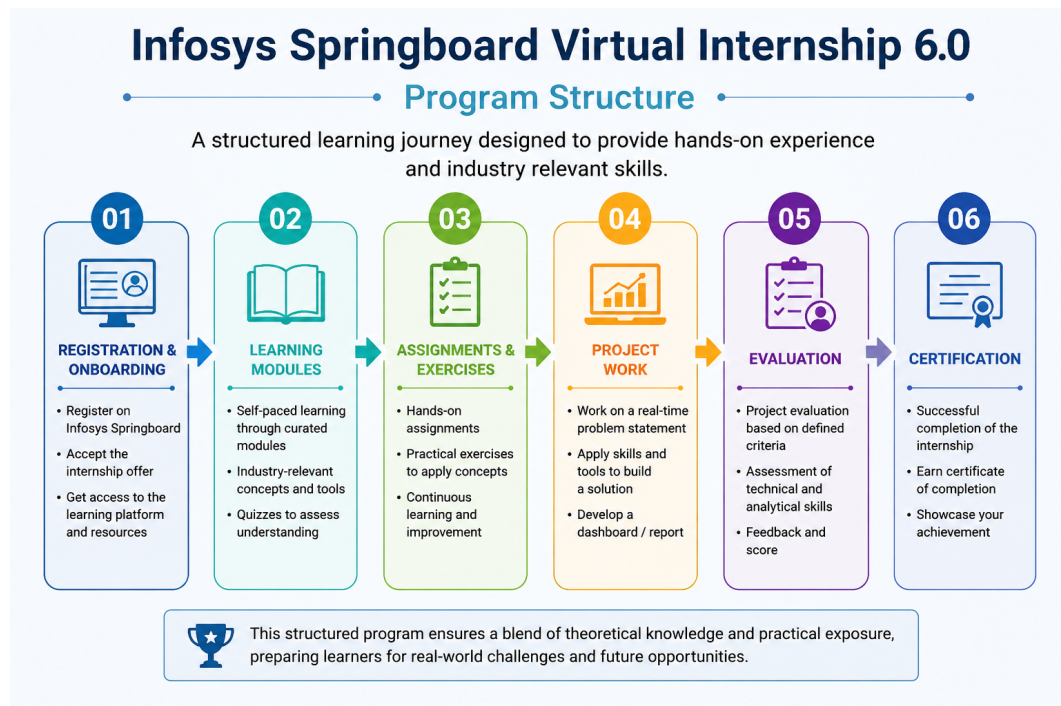


Fig 2.2: Virtual Internship 6.0 Program Structure

riving actionable business insights across multiple analytical dimensions. The project was completed in structured milestones, with each milestone representing a significant stage of progress in the overall dashboard development. Throughout the project phase, students received regular feedback and guidance from experienced Infosys mentors who reviewed milestone submissions and provided constructive suggestions to help students continuously improve the quality and depth of their work. This mentorship element is one of the most valuable aspects of the program, exposing students to the kind of professional feedback and iterative improvement process that is central to real-world project development in the technology industry. The program concludes with a final submission and evaluation phase, during which students submit their completed project for comprehensive assessment. The evaluation criteria assess not only the technical quality and completeness of the deliverable but also the depth of analytical insight, the clarity and effectiveness of the visual communication, the coherence of the data story, and the overall professionalism of the presentation. Students who successfully meet the evaluation standards receive an official internship completion certificate from Infosys Springboard, which is widely recognized by employers as a credible and meaningful indicator of industry-relevant skills, professional commitment, and the ability to deliver results in a structured and deadline-driven environment. The Infosys Springboard Virtual Internship 6.0 program thus serves as a powerful launchpad for students entering the technology workforce, equipping them with the practical experience, technical competence, and professional confidence needed to make a strong and impactful start to their careers.

TECHNOLOGY STACK

The successful development of the Food Trends: Understanding Customer Preferences in Food and Beverages dashboard required the use of a carefully selected set of tools and technologies that together formed a robust, efficient, and industry-standard technology stack. Each tool was chosen for its specific capabilities, its relevance to the data visualization domain, and its suitability to the analytical and design requirements of the project. The primary technology used throughout the internship was Microsoft Power BI, supported by Power Query for data connectivity and transformation, DAX for analytical calculations and KPI development, and a carefully structured data modeling approach that enabled seamless cross-dimensional analysis across the entire dataset. Together, these technologies provided a comprehensive and powerful environment for building a professional-grade, multi-page interactive dashboard that delivers actionable business insights from a complex real-world dataset.

This chapter provides an in-depth and comprehensive overview of each technology used during the internship, explaining what it is, its history and evolution, its key features and capabilities, its advantages and industry relevance, why it was selected for this project, and how it was specifically and practically applied in the context of the Food Trends dashboard. Understanding the technology stack in depth is essential for appreciating the technical rigor and analytical sophistication of the work undertaken during the internship, and for recognizing how each individual tool contributed to the overall quality, accuracy, interactivity, and professional presentation of the final deliverable. The technology stack discussed in this chapter represents not only the tools used in this specific project but also the core foundational toolkit of any modern data visualization and business intelligence professional working in today's technology industry.

3.1 Microsoft Power BI

Microsoft Power BI is a world-leading business intelligence and data visualization platform developed by Microsoft Corporation that enables users to connect to a wide variety of data sources, transform and model data, and create interactive, visually rich reports and dashboards that communicate insights clearly and effectively to both technical and non-technical audiences. First introduced by Microsoft in 2011 and officially launched as a standalone product in 2015, Power BI has rapidly grown to become one of the most widely adopted business intelligence tools in the world, consistently recognized as a Leader in the Gartner Magic Quadrant for Analytics and Business Intelligence Platforms. It is used by organizations of all sizes across virtually every industry, from small businesses leveraging it for operational reporting to large multinational corporations using it for enterprise-wide strategic analytics and executive decision-making.

Power BI is available in multiple versions and deployment options to accommodate different organizational needs and use cases. Power BI Desktop is a free Windows application used by report developers and data analysts to build and design reports and dashboards. Power BI Service is a cloud-based software-as-a-service platform hosted on Microsoft Azure that allows users to publish, share, and collaborate on Power BI reports and dashboards within an organization. Power BI Mobile is a suite of native mobile applications for iOS, Android, and Windows devices that enables users to access and interact with their Power BI reports on smartphones and tablets. Power BI Premium and Power BI Embedded are enterprise-grade offerings that provide enhanced performance, capacity, and integration capabilities for organizations with large-scale business intelligence requirements.



Fig 3.1: Microsoft Power BI

Power BI was selected as the primary tool for this internship project for several compelling reasons. First, it is the industry’s leading business intelligence platform and is widely used by data professionals across the globe, making proficiency in Power BI a highly valuable and sought-after skill in the job market. Second, Power BI offers an exceptionally powerful combination of ease of use and analytical depth, allowing users to build sophisticated and visually compelling dashboards without requiring extensive programming knowledge, while still supporting advanced analytical capabilities through DAX and Power Query for users who need them. Third, Power BI’s rich library of visualization types — including bar charts, column charts, line charts, area charts, pie and donut charts, scatter plots, bubble charts, maps, filled maps, heatmaps, treemaps, waterfall charts, funnel charts, gauge visuals, card visuals, multi-row card visuals, slicers, tables, matrices, and a growing collection of AI-powered visuals such as the Key Influencers visual, Decomposition Tree, Smart Narrative, and Q&A visual — provided all the visualization capabilities needed to address the diverse analytical requirements of the Food Trends project across its seven report pages. During the internship, Power BI Desktop was used as the primary development environment for the entire project. It was used to connect to and import the Food Trends dataset, apply data transformation steps in Power Query, define the data model and table relationships, write DAX measures and calculated columns, and design and develop all seven pages

of the interactive dashboard. Power BI's intuitive drag-and-drop report canvas made it efficient to build, iterate, and refine visual layouts, while its extensive formatting options and theming capabilities — including custom color palettes, font selections, background settings, border and shadow effects, and conditional formatting — ensured that the final dashboard was visually consistent, professionally presented, and aesthetically appealing. The interactive filtering, cross-highlighting, and drill-through capabilities of Power BI transformed the dashboard from a static report into a truly dynamic analytical tool, enabling users to explore data from multiple perspectives and at different levels of granularity with intuitive and responsive interactions.

3.2 DAX (Data Analysis Expressions)

DAX, which stands for Data Analysis Expressions, is the formula language used in Microsoft Power BI, as well as in Power Pivot for Excel and SQL Server Analysis Services, to define custom calculations, measures, and calculated columns within a data model. Introduced by Microsoft in 2009 as part of Power Pivot for Excel, DAX has evolved into a sophisticated and expressive analytical language that is central to the analytical power of any Power BI implementation. DAX is modeled after Excel formula syntax, making it relatively accessible to users with a spreadsheet background, while offering a far greater level of power, flexibility, and performance than Excel formulas, particularly when working with large datasets and complex multi-table data models.



Fig 3.2: DAX (Data Analysis Expressions)

DAX operates on the principle of filter context and row context, which are the two fundamental evaluation contexts that determine how a DAX formula calculates its result at any given point in a report. Filter context refers to the set of filters applied to the data model by report visuals, slicers, and page filters, which determine which rows of data are included in a calculation at any given moment. Row context refers to the current row being evaluated in an iterative calculation, such as a calculated column or an iterator function. Understanding and correctly applying these two evaluation contexts is the key to writing accurate and effective DAX measures, and mastering this concept was one

of the most intellectually challenging and rewarding aspects of the technical learning during the internship.

DAX was used extensively throughout the Food Trends project to create the calculated measures and KPIs that powered the dashboard's analytical insights across all seven report pages. The measures developed using DAX included Total Orders, which aggregates the total count of orders in the dataset, Total Restaurants, which counts the distinct number of restaurants, Total Cities, which counts the number of unique cities represented in the data, Average Rating, which calculates the mean customer rating across all restaurants, Average Price, which computes the mean price across all records, Average Delivery Time, which measures the mean delivery duration, Average Order Value, which calculates the mean revenue per order, Fast Delivery Percentage, which computes the proportion of orders delivered within the target delivery time, Delivery Performance Score, which provides a composite measure of delivery efficiency, Positive Restaurant Count, Negative Restaurant Count, and Neutral Restaurant Count, which segment restaurants by their dominant sentiment classification, Top Performing City, which identifies the city with the highest combined performance score, and Dominant Satisfaction Level, which identifies the most frequently occurring customer satisfaction category across the dataset.



Fig 3.3: DAX Functions

Writing these DAX measures required a deep understanding of the data model, the relationships between tables, and the filter context in which each measure would be evaluated. A wide range of DAX functions were used across the project, including aggregation functions such as SUM, COUNT, COUNTROWS, DISTINCTCOUNT, AVERAGE, MIN, and MAX for basic metric calculations, iterator functions such as SUMX, AVERAGEX, and COUNTX for row-by-row calculations over a table, filter manipulation functions such as CALCULATE, FILTER, ALL, ALLEXCEPT, and RE-

MOVEFILTERS for controlling and modifying the filter context of calculations, logical functions such as IF, SWITCH, AND, and OR for conditional logic, text functions such as CONCATENATE and FORMAT for creating dynamic labels and descriptions, ranking functions such as RANKX for ordering and ranking entities by performance metrics, and mathematical functions such as DIVIDE for safe division that handles zero denominators gracefully. The ability to write sophisticated and accurate DAX measures was one of the most technically demanding aspects of the project and one of the most valuable and transferable skills developed during the internship.

3.3 Power Query

Power Query is a powerful data connectivity and transformation engine built into Microsoft Power BI, as well as into Microsoft Excel and other Microsoft data products, that enables users to import data from a wide variety of sources, clean and reshape it through a series of applied transformation steps, and load it into the data model in a structured, consistent, and analysis-ready format. Power Query uses an intuitive graphical, step-based interface known as the Power Query Editor, which records every transformation applied to the data as a discrete, named step in an applied steps pane, making the entire data preparation process fully transparent, reproducible, and easy to audit, review, or modify at any point during development. Each step in the applied steps sequence is automatically translated into M code, the underlying functional programming language that powers Power Query, which can also be written directly by advanced users who need to implement custom transformation logic that goes beyond what the graphical interface supports. Power Query supports connectivity to an exceptionally wide range of data sources, including flat files such as CSV, Excel, and JSON files, relational databases such as SQL Server, MySQL, PostgreSQL, and Oracle, cloud data services such as Azure SQL Database, Google BigQuery, and Amazon Redshift, online services such as SharePoint, Salesforce, and Dynamics 365, web pages, OData feeds, and many more. This broad connectivity makes Power Query a highly versatile data preparation tool that can be applied in virtually any data environment, regardless of where the source data resides or in what format it is stored.

In the context of the Food Trends project, Power Query was used to import the raw dataset and perform a comprehensive series of data cleaning and transformation operations to ensure that the data was accurate, consistent, complete, and suitable for analysis. The transformation process began with a thorough inspection of the raw data to identify quality issues, inconsistencies, and anomalies that needed to be addressed before analysis could begin. The specific transformation steps applied during the project included removing duplicate records that could distort aggregated metrics, identifying and appropriately handling null and missing values across key columns, correcting data



Fig 3.4: Power Query

type assignments to ensure that numeric columns were recognized as numbers, date columns were recognized as dates, and categorical columns were treated as text, renaming columns with clearer and more descriptive names to improve the readability and usability of the data model, trimming and cleaning text fields to remove leading and trailing whitespace and other formatting inconsistencies, creating new conditional columns to derive calculated categorical variables such as Price Segment, which classified restaurants into Low, Mid-Range, and Premium categories based on their average price, Rating Segment, which classified restaurants into Poor, Average, Good, and Excellent categories based on their average customer rating, Delivery Segment, which classified orders by delivery speed into Fast, Standard, and Slow categories, and Sentiment classification, which assigned a Positive, Neutral, or Negative sentiment label based on rating thresholds. These derived categorical columns were essential for enabling the segmentation and classification analyses that form a core part of the dashboard's analytical framework. The application of Power Query during this project reinforced the fundamental principle that data quality is the foundation of analytical credibility. Every insight derived from the dashboard, every KPI displayed on a card visual, and every trend visualized in a chart is only as reliable as the underlying data from which it is calculated. By investing significant effort in the data preparation phase through Power Query, the integrity and trustworthiness of all subsequent analysis was assured, and the risk of misleading or inaccurate insights was minimized.

3.4 Data Modelling

Data modeling is the process of defining the structure, organization, and relationships of data within a business intelligence system to ensure that it can be queried efficiently, accurately, and flexibly in support of the analytical requirements of a report or dashboard. In Microsoft Power BI, data modeling is performed in the Model view, where tables imported from Power Query are connected through defined relationships, and where additional structural elements such as hierarchies, calculated columns, and

measures are created to extend the analytical capabilities of the model. A well-designed data model is the single most important architectural foundation of any effective Power BI implementation, as it determines how data from different tables can be combined and analyzed together, how DAX measures filter and aggregate data in response to user interactions, and whether the report delivers accurate and consistent results across all its visuals and analytical scenarios. Data modeling in Power BI is built on the principles of dimensional modeling, a well-established methodology in the business intelligence field that organizes data into fact tables and dimension tables. Fact tables contain the quantitative, transactional data that is the subject of analysis — in this case, the Fact Orders table containing order-level records with associated metrics such as revenue, delivery time, ratings, and order frequency. Dimension tables contain the descriptive, categorical attributes that provide context for the facts — such as restaurant names, cuisines, cities, price segments, and rating segments. Relationships between fact tables and dimension tables are defined in the data model, allowing Power BI to combine and filter data across tables in response to user interactions with slicers, filters, and visual selections.

For the Food Trends project, the data model was built around the central Fact Orders table, which served as the primary source of all transactional and behavioral data in the dataset. Since the project utilized a single unified dataset, the data model was structured to maximize the analytical flexibility of this single table by creating a rich set of calculated columns and DAX measures that extended its analytical capabilities across multiple dimensions simultaneously. The data model was designed with careful attention to ensure that all analytical dimensions — including customer behavior, product and cuisine preferences, sentiment classification, delivery performance, pricing analysis, and geographic distribution — could be analyzed both independently and in meaningful combination with each other across all seven report pages.



Fig 3.5: Data Modeling

Relationships between tables were carefully defined with the correct cardinality settings — one-to-many, many-to-one, or many-to-many as appropriate — and the correct

cross-filter direction to ensure accurate and consistent results across all visuals and measures throughout the report. Hierarchies were created within the data model to enable intuitive drill-down analysis within geographic dimensions, allowing users to explore data progressively from the city level down to the restaurant level, and within categorical dimensions such as price and rating segments, enabling analysis at different levels of granularity. The data model also incorporated a number of calculated columns created using DAX that extended the base dataset with derived variables needed for segmentation, classification, and ranking analyses, such as customer engagement categories, delivery performance classifications, and composite satisfaction scores.

The overall data modeling approach adopted for the Food Trends project reflects industry best practices in Power BI development and ensures that the dashboard is not only analytically accurate and reliable today but is also structured in a way that would support future enhancements, additional data sources, and evolving analytical requirements without requiring a fundamental redesign of the underlying model. The discipline of data modeling — understanding how data is structured, how tables relate to each other, and how analytical calculations flow through the model — is one of the most foundational and transferable skills in the field of business intelligence, and developing this competence during the internship has provided a strong and lasting foundation for future work in data analytics and visualization.

TECHNICAL TRAINING AND ASSIGNMENT

The technical training phase of the Infosys Springboard Virtual Internship 6.0 was one of the most structured, comprehensive, and intellectually enriching components of the entire internship experience. Before being assigned to work on the Food Trends project, all interns in the Data Visualization domain were required to complete a carefully designed and progressively structured training program that covered the full spectrum of knowledge and skills needed to independently design, develop, and deliver a professional-grade Power BI dashboard. The training was not merely an introductory orientation but a thorough and rigorous skill-building program that took participants from foundational concepts all the way through to advanced analytical and design techniques, ensuring that every intern was fully equipped and confidently prepared to tackle the demands of a real-world data visualization project.

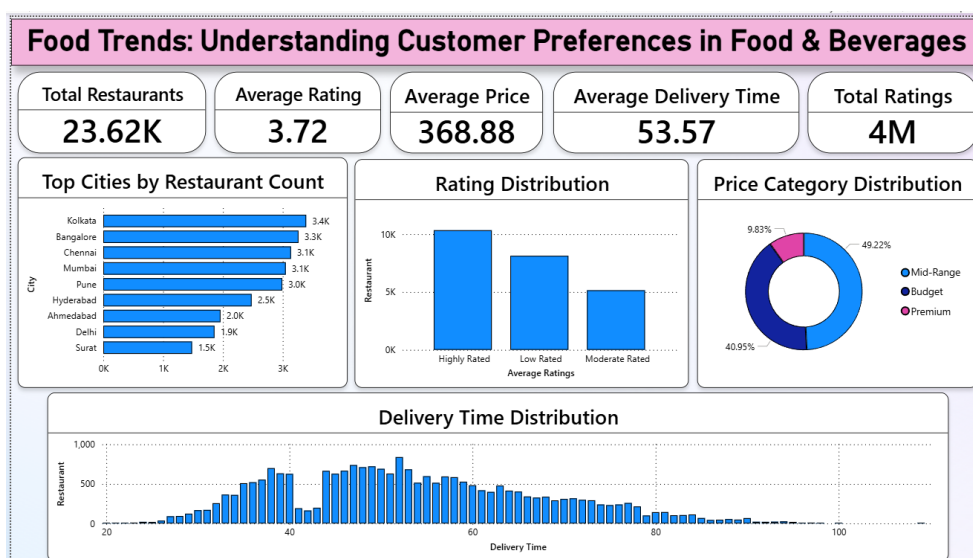


Fig 4.1: Sample dashboard created during Training Phase

The training program was delivered through the Infosys Springboard platform in the form of structured learning modules, each covering a specific topic or technology area relevant to the data visualization domain. Each module combined conceptual explanations with practical demonstrations, worked examples, and hands-on exercises that required learners to actively apply what they had learned rather than passively consuming content. This active learning approach was central to the effectiveness of the training, as it ensured that knowledge was not merely understood in theory but was genuinely internalized and practically applicable. The training was organized into several key areas including data visualization principles and best practices, Microsoft Power BI fundamentals and advanced features, data transformation and preparation using Power

Query, data modeling concepts and implementation, DAX formula writing and measure development, dashboard design and layout principles, and data storytelling techniques. Each of these areas is discussed in detail in the sections that follow.

4.1 Data Visualization Principles

The training program began with a thorough grounding in the fundamental principles of data visualization — the science and art of representing data graphically in ways that communicate information clearly, accurately, and effectively to the intended audience. This foundational phase was essential because effective data visualization is not simply a matter of creating charts and graphs; it requires a deep understanding of how human beings perceive and process visual information, what types of visual representations are most appropriate for different kinds of data and analytical questions, and how design choices such as color, layout, typography, and visual hierarchy influence the clarity and impact of a dashboard. During this phase of the training, I learned about the core principles that distinguish effective data visualization from poor or misleading visual communication. The principle of clarity emphasizes that every visual element in a dashboard should serve a clear and purposeful communicative function, and that unnecessary complexity, decoration, or visual noise should be eliminated to ensure that the audience can focus on the data and the insights it conveys. The principle of accuracy requires that visual representations faithfully and proportionally represent the underlying data, avoiding distortions such as truncated axes, inconsistent scales, or misleading chart types that could cause audiences to draw incorrect conclusions. The principle of efficiency holds that a well-designed visualization should enable the audience to extract the key insight as quickly and effortlessly as possible, minimizing the cognitive effort required to understand what the data is saying.

I also learned about the importance of choosing the right chart type for the right analytical purpose. Bar and column charts are most effective for comparing discrete categories. Line charts are best suited for showing trends over time or continuous variables. Pie and donut charts are appropriate only for showing part-to-whole relationships with a small number of categories. Scatter charts are ideal for revealing correlations and distributions between two continuous variables. Maps are the natural choice for geographic data. Card visuals are most effective for prominently displaying single key metrics. Heatmaps are powerful for showing the intensity of a variable across two dimensions simultaneously. Understanding these principles of chart selection and applying them consistently throughout the Food Trends dashboard was fundamental to ensuring that every visualization on every page communicated its intended insight as clearly and effectively as possible. The training also covered the principles of visual hierarchy and layout design, explaining how the arrangement and relative sizing of visual elements

on a dashboard page guides the audience's eye and communicates the relative importance of different pieces of information. High-priority KPIs and summary metrics are typically placed at the top of a dashboard page, where they are seen first, while supporting charts and detailed breakdowns are arranged below in a logical reading order that guides the audience through the data story progressively. Color theory was another important topic covered during this phase, including how to use color purposefully and consistently to encode information, highlight key insights, distinguish categories, and maintain visual harmony across a dashboard without overwhelming the audience with excessive color variation.

4.2 Power BI Fundamentals

Following the grounding in data visualization principles, the training moved into a comprehensive introduction to Microsoft Power BI, beginning with the fundamentals of the platform and progressively advancing to more sophisticated features and techniques. This phase of the training was the most extensive and practically intensive component of the program, as Power BI was the primary tool used for all project work and required a thorough understanding of its full feature set to use effectively. The Power BI fundamentals training began with an orientation to the Power BI Desktop interface and its key components. The Report view is where reports and dashboards are designed and developed, providing a visual canvas on which visuals can be placed, sized, formatted, and arranged. The Data view provides a tabular representation of the data loaded into the model, allowing users to inspect the data, verify transformations, and review calculated columns. The Model view provides a visual representation of the data model, showing the tables, columns, and relationships that form the analytical foundation of the report. The Fields pane lists all the tables, columns, calculated columns, and measures available in the data model, from which they can be dragged onto the report canvas to create visuals. The Visualizations pane provides access to all available visualization types and their formatting options. Understanding the purpose and functionality of each of these interface components was the essential first step in becoming productive in Power BI Desktop. The training then progressed to cover the process of connecting Power BI to data sources and importing data. Power BI supports connectivity to an exceptionally wide range of data sources through its Get Data functionality, including flat files such as CSV and Excel files, relational databases such as SQL Server and MySQL, cloud services such as Azure and Google BigQuery, and many others. For the Food Trends project, the dataset was provided as a structured flat file that was imported into Power BI Desktop using the Get Data functionality, initiating the data preparation workflow in Power Query.

The training also covered the creation and configuration of visualizations in Power

BI in considerable depth. I learned how to add visuals to the report canvas by selecting them from the Visualizations pane, how to assign data fields to the appropriate visual wells such as Axis, Values, Legend, and Tooltips to control what data each visual displays, and how to use the Format pane to configure every aspect of a visual's appearance including colors, fonts, borders, backgrounds, data labels, axis titles and formatting, legend position and formatting, and many other properties. I also learned how to use slicers to enable interactive filtering, how to configure cross-highlighting and cross-filtering behavior between visuals, how to add and configure drill-through pages, and how to implement page-level, report-level, and visual-level filters to control what data is displayed in different parts of the report. Advanced Power BI features covered during the training included the use of bookmarks and selection panes to create interactive navigation experiences, the implementation of buttons and navigation actions to enable users to move between report pages, the use of conditional formatting to dynamically color visual elements based on data values, the creation and application of custom themes to ensure visual consistency across all pages of a report, the use of field parameters to enable dynamic axis and measure selection by report users, and the integration of AI-powered visuals such as the Key Influencers visual and the Decomposition Tree into reports to deliver machine learning-enhanced analytical insights without requiring any programming or data science expertise.

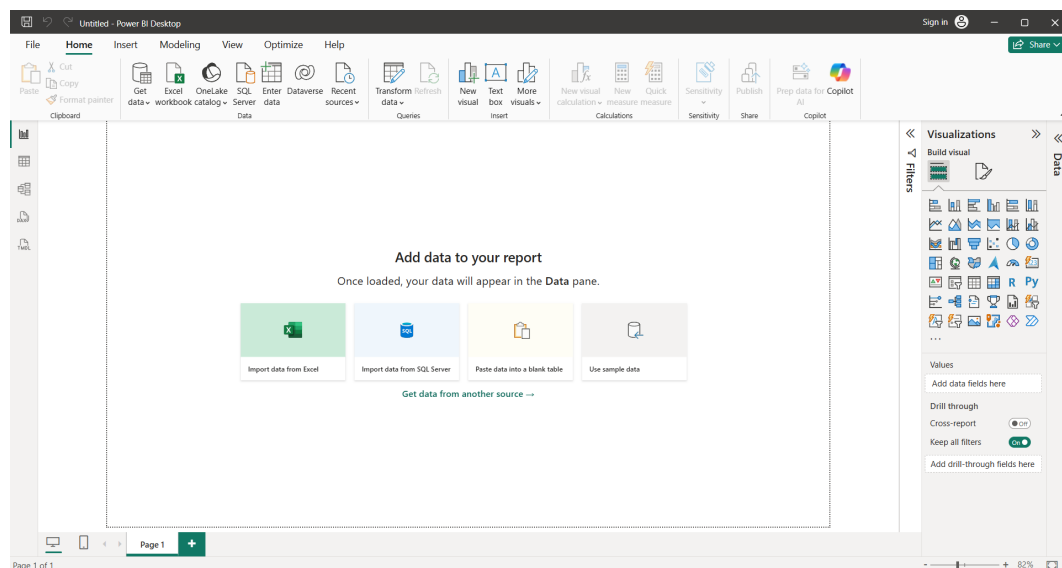


Fig 4.2: Power BI Desktop Interface

The above figure shows the Microsoft Power BI Desktop interface used for developing the dashboard in this project. The Report View provides a canvas where various visualizations such as charts, graphs, and KPI cards are created and arranged. The Fields Pane displays the dataset columns and measures, which can be dragged and dropped to build visuals. The Visualizations Pane contains different chart types and formatting options that help in designing interactive dashboards. The Filters Pane allows users

to apply filters at visual, page, and report levels to refine the analysis. This interface played a crucial role in creating and customizing the Food Trends dashboard.

4.3 DAX Formula Writing

The DAX formula writing module was one of the most technically challenging and intellectually stimulating components of the training program. DAX, which stands for Data Analysis Expressions, is the formula language used in Power BI to create calculated columns and measures that extend the analytical capabilities of the data model beyond what can be achieved with simple aggregations of raw data fields. Developing proficiency in DAX is widely regarded as the single most important technical skill that distinguishes a basic Power BI user from a genuinely capable and effective Power BI developer, as the quality and sophistication of the DAX measures in a report directly determines the depth, accuracy, and analytical power of the insights it delivers. The DAX training began with an explanation of the two fundamental types of DAX calculations. Calculated columns are DAX formulas that are evaluated row by row during data refresh and stored as new columns in a table within the data model. They are used to create derived attributes and classifications that extend the base dataset, such as categorizing restaurants into price segments or rating segments based on their raw numeric values. Measures are DAX formulas that are evaluated dynamically at query time in response to the filter context established by the current state of the report, including the filters applied by slicers, page filters, and visual interactions. Measures are not stored as data in the model but are calculated on demand, making them the primary mechanism for creating KPIs, aggregated metrics, and complex analytical calculations in Power BI. The training covered the concept of filter context in depth, as it is the most important and often the most challenging concept to master in DAX. Filter context refers to the set of filters that are active at the point where a measure is being evaluated, which determines which rows of data are included in the calculation. Every visual in a Power BI report automatically establishes a filter context based on the fields assigned to it and the filters applied to it, and DAX measures respond to this filter context to produce results that are specific to the data visible in that particular visual at that particular moment. Understanding how filter context works, how it flows through the data model along relationship paths, and how it can be modified, expanded, or removed using DAX functions such as `CALCULATE`, `ALL`, `ALLEXCEPT`, `REMOVEFILTERS`, and `KEEPFILTERS` was one of the most important and impactful areas of learning during the training program.

The specific DAX functions covered during the training included a comprehensive range of the most important and commonly used functions in the DAX language. Aggregation functions including `SUM`, `COUNT`, `COUNTROWS`, `DISTINCTCOUNT`,

AVERAGE, MIN, MAX, and MEDIAN for computing basic summary statistics. Iterator functions including SUMX, AVERAGEX, COUNTX, MAXX, and MINX for performing row-by-row calculations over a specified table and then aggregating the results. The CALCULATE function, which is arguably the single most important function in DAX, used to evaluate a measure or expression in a modified filter context by adding, removing, or overriding filters. Filter functions including FILTER, ALL, ALLEXCEPT, ALLSELECTED, REMOVEFILTERS, and KEEPFILTERS for controlling and manipulating the filter context of calculations. Logical functions including IF, SWITCH, AND, OR, NOT, IFERROR, and ISBLANK for implementing conditional logic and error handling. Text functions including CONCATENATE, CONCATENATEX, FORMAT, LEFT, RIGHT, MID, LEN, UPPER, LOWER, and TRIM for text manipulation and dynamic label creation. Mathematical functions including DIVIDE, ROUND, ROUNDUP, ROUNDDOWN, ABS, INT, MOD, POWER, SQRT, and CEILING for numerical calculations and formatting. Ranking and statistical functions including RANKX, PERCENTILE, STDEV, VAR, TOPN, and GENERATE for advanced analytical calculations. Time intelligence functions including TOTALYTD, TOTALQTD, TOTALMTD, DATEADD, SAMEPERIODLASTYEAR, DATESYTD, and PREVIOUSMONTH for time-based calculations, which while not the primary focus of the Food Trends project provided important foundational knowledge for future analytical work.

4.4 Power Query Transformations

The Power Query training module covered the complete workflow of data preparation in Power BI, from initial data connection and import through to the full range of transformation operations needed to produce a clean, consistent, and analysis-ready dataset. Power Query is the data transformation engine built into Power BI that provides a graphical, step-based interface for applying transformations to imported data, recording each transformation as a discrete step in the applied steps sequence that can be reviewed, modified, reordered, or deleted at any time during development. The training began with an introduction to the Power Query Editor interface and its key components, including the ribbon of transformation commands, the applied steps pane that shows the sequence of transformations applied to the data, the data preview pane that shows the current state of the data after all applied steps have been executed, the queries pane that lists all the queries in the Power Query session, and the formula bar that shows the M code generated by each applied step. Understanding how to navigate and work effectively within this interface was the essential foundation for all subsequent data preparation work.

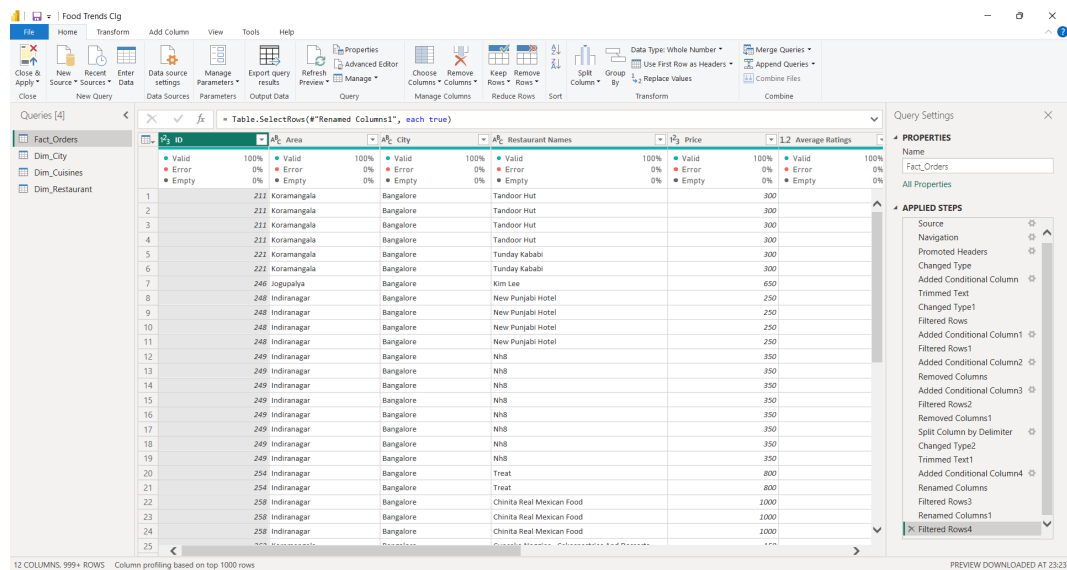


Fig 4.3: Power Query Editor

The above figure shows the Power Query Editor used during the data preparation phase of the Food Trends Analysis project. Power Query was utilized to clean, transform, and organize the raw dataset before loading it into Power BI. Various data preprocessing operations such as changing data types, trimming text values, removing unnecessary columns, filtering records, splitting columns, and creating conditional columns were performed to improve data quality and consistency. The Applied Steps pane on the right displays the sequence of transformations executed on the dataset, ensuring a structured and reproducible data cleaning process. These transformations helped eliminate data inconsistencies, improve accuracy, and prepare the data for effective modelling, analysis, and dashboard development.

The transformation operations covered during the training included the full range of data cleaning and reshaping tasks commonly encountered in real-world data preparation workflows. Removing duplicate rows to eliminate redundant records that could distort aggregated metrics. Removing errors and null values or replacing them with appropriate default values or interpolated estimates to ensure completeness and consistency. Changing data types to ensure that each column is assigned the correct data type — numeric, text, date, boolean, or other — so that Power BI can perform appropriate operations on each column. Renaming columns and queries to use clear, descriptive, and consistent naming conventions that improve the readability and maintainability of the data model. Filtering rows based on specified criteria to remove irrelevant, out-of-scope, or erroneous records from the dataset. Adding conditional columns using if-then-else logic to create new categorical variables derived from existing numeric or text columns. Splitting columns by delimiter or by position to separate combined fields into individual components. Merging columns to combine multiple fields into a single composite field. Grouping and aggregating data to summarize records at a higher

level of granularity. Pivoting and unpivoting columns to reshape data between wide and long formats. Merging queries to combine data from multiple tables based on matching key fields, similar to a SQL JOIN operation. Appending queries to combine data from multiple tables with the same structure into a single unified table.

The training also introduced the M language, the functional programming language that underlies Power Query and that is automatically generated as users apply transformations through the graphical interface. While the graphical interface handles the vast majority of common transformation scenarios without requiring any manual M coding, understanding the basics of M syntax and structure enables more advanced users to write custom transformation logic, create reusable functions, and implement transformations that are not available through the graphical interface alone. During the training, I gained a foundational understanding of M syntax including the `let` expression structure, the `in` keyword, the use of step references to chain transformations, and the application of common M functions for text manipulation, list operations, and conditional logic.

4.5 Dashboard Design and Data Storytelling

The dashboard design and data storytelling module addressed the visual and communicative dimensions of Power BI report development, complementing the technical training in Power Query and DAX with guidance on how to create dashboards that are not only analytically accurate but also visually compelling, intuitively navigable, and effective at communicating insights to their intended audience. This module was grounded in the recognition that a Power BI dashboard is ultimately a communication tool, and that its value is determined not only by the quality of the underlying data and calculations but equally by how effectively the visual design enables the audience to understand and act on the insights it presents. The training covered the principles of effective dashboard layout and composition, emphasizing the importance of establishing a clear visual hierarchy that guides the audience's attention from the most important summary information at the top of the page to the supporting detail and contextual analysis below. I learned about the rule of proximity, which holds that related visual elements should be placed close together to signal their relationship, and the rule of alignment, which requires that visual elements be consistently aligned to create a sense of order and professionalism. The use of white space — the empty areas between and around visual elements — was discussed as an important design tool that prevents dashboards from feeling cluttered and overwhelming, and that helps to define visual groupings and direct the audience's focus. Color design was covered in considerable depth during this module, including the principles of using a limited and consistent color palette, choosing colors that are meaningful and encode information rather than merely decorative,

ensuring sufficient contrast between text and background colors for readability and accessibility, and using color strategically to highlight the most important data points and guide the audience's attention. The training also addressed the specific considerations involved in designing dashboards for users with color vision deficiencies, emphasizing the importance of not relying on color alone to encode information and of testing dashboard designs against colorblind simulation tools to verify their accessibility.

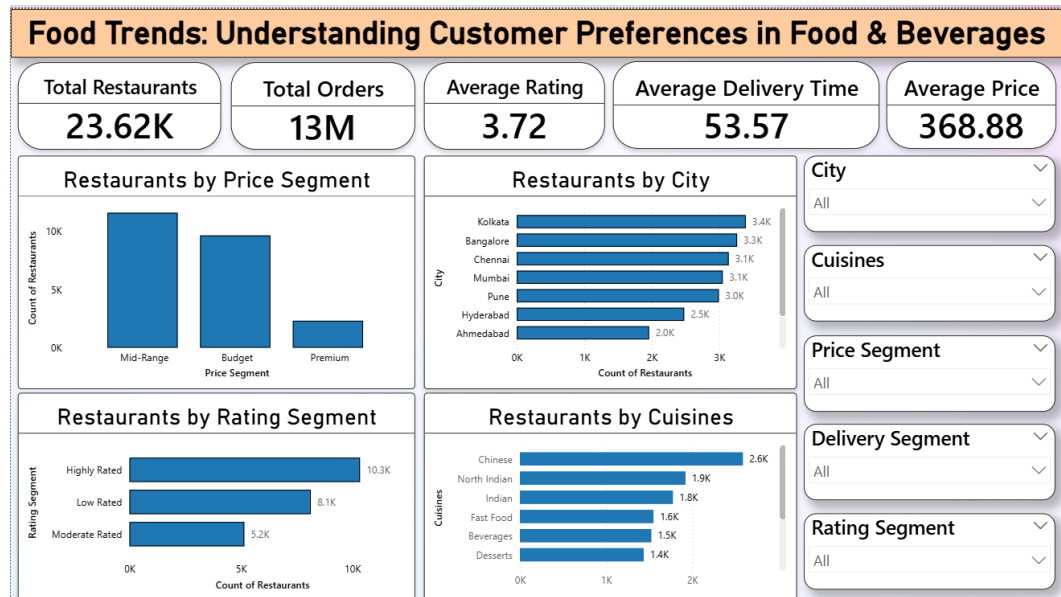


Fig 4.4: Data Storytelling Dashboard

Data storytelling was presented as the overarching framework within which all dashboard design decisions should be made. A data story is a structured narrative that uses data visualizations to take the audience on a logical, coherent journey from a statement of the business context and the analytical questions being addressed, through the evidence provided by the data, to the insights and conclusions that the data supports. I learned how to structure a multi-page Power BI report as a data story by organizing the pages in a logical narrative sequence, with each page building on the insights of the previous one and contributing a distinct and complementary analytical perspective to the overall story. For the Food Trends project, this data storytelling framework was applied by organizing the seven report pages in a sequence that begins with the high-level executive summary, moves through increasingly specific analytical perspectives on customer sentiment, product preferences, segmentation, and strategy, and concludes with the operational and geographic analyses that provide the most granular and actionable insights.

4.6 Training Assignments and Exercises

Throughout the training program, learners were required to complete a carefully designed and progressively structured series of practical assignments and exercises specif-

ically crafted to reinforce the concepts and skills covered in each training module and to develop genuine hands-on proficiency with the tools and methodologies being taught. These assignments were not merely theoretical assessments or quizzes designed to test recall of information. Rather, they were substantive practical tasks that required learners to work directly with real or simulated datasets in Power BI Desktop, apply the skills taught in the corresponding training modules to accomplish defined analytical and design objectives, and produce tangible deliverables that demonstrated both their technical understanding and their ability to make sound analytical and design judgments in realistic data visualization scenarios. The assignments were structured to progress systematically in complexity and scope, following a carefully designed learning arc that began with focused, single-skill exercises in the earliest stages of the training and gradually advanced to comprehensive, multi-skill integration tasks as learners' capabilities developed. This scaffolded progression was designed to build learner confidence progressively, ensuring that each new level of complexity was introduced only after the foundational skills it depended upon had been sufficiently practiced and internalized, mirroring the natural workflow of a real Power BI development project. The earliest assignments targeted individual competencies within the Power BI workflow. Data connectivity exercises required learners to practice connecting Power BI Desktop to different types of data sources and successfully importing data into the Power Query Editor. Power Query transformation exercises provided learners with intentionally messy datasets containing common data quality issues — including duplicate records, null values, incorrect data types, and inconsistently formatted text fields — and required them to apply the appropriate transformation steps to clean, standardize, and reshape the data into a form suitable for analysis. These exercises progressed from simple single-step transformations through to multi-step sequences that required learners to chain together a series of operations to produce a fully cleaned and restructured dataset from a raw source file.

The DAX formula writing exercises were among the most technically demanding assignments in the training program. These exercises began with the creation of simple calculated columns and basic aggregation measures, and progressively introduced more complex DAX concepts including conditional logic using IF and SWITCH, filter context manipulation using CALCULATE combined with ALL and ALLEXCEPT, percentage contribution calculations, ranking measures using RANKX, and top performer identification using TOPN and FILTER. Learners were also assigned comprehensive measure-writing tasks that required them to develop a complete suite of interrelated measures for a given dataset, test each measure by verifying its output across different filter contexts, and organize the measures logically within the data model using display folders. The visualization design exercises focused on developing learners' ability to select appropriate chart types for different analytical purposes, configure and format

visuals to communicate insights clearly and professionally, and arrange visuals on a report canvas following sound dashboard design principles. One of the most valuable design exercises required learners to take a poorly designed dashboard and completely redesign it by applying the principles of effective data visualization covered in the training, developing critical design thinking by identifying what was wrong with the existing design and articulating why the changes made represented genuine improvements.

The culminating assignment of the training program was a comprehensive mini-project that required learners to independently design and develop a complete multi-page Power BI dashboard from scratch using a provided sample dataset, applying the full range of skills acquired across all training modules. The mini-project began with an exploratory data analysis phase to understand the dataset's structure, quality issues, and analytical potential, followed by defining a clear analytical objective, planning the report structure, and then executing the full development workflow from data preparation in Power Query through data modeling and DAX measure creation to the design and formatting of all report pages. The completed mini-project was submitted for review and feedback by program mentors, who evaluated it against criteria covering data preparation quality, data model design, DAX measure accuracy, visualization appropriateness, dashboard layout, interactivity, and the overall coherence of the data story. The detailed and constructive feedback received from mentors was carefully reviewed and thoroughly incorporated into the approach taken for the main Food Trends project, making the mini-project an integral and formative step in the development of the professional standards and capabilities applied in the final project deliverable.

INTERNSHIP PROJECT CONTRIBUTION

The internship project contribution phase represented the most significant, demanding, and professionally rewarding component of the entire Infosys Springboard Virtual Internship 6.0 experience. Having successfully completed the structured training program that equipped me with the foundational and advanced skills required for data visualization work in Microsoft Power BI, I transitioned into the active project development phase where all the knowledge, techniques, and professional practices developed during training were applied to a real-world analytical challenge. The project assigned to me within the Data Visualization domain was the development of a comprehensive, multi-page interactive dashboard titled "Food Trends: Understanding Customer Preferences in Food and Beverages" — a business intelligence solution designed to analyze customer behavior, sentiment patterns, product preferences, delivery performance, and geographic market dynamics across major Indian cities using a rich and complex real-world dataset sourced from one of India's most prominent food delivery platforms.

The project required me to independently manage and execute the complete end-to-end workflow of a professional Power BI development project, from the initial sourcing, exploration, and preparation of the raw dataset through the design and construction of the data model, the development of the full suite of DAX measures and KPIs, and the design, formatting, and refinement of all seven analytical report pages that make up the final dashboard. Throughout the project phase, I applied not only the technical skills developed during training but also the professional disciplines of structured project planning, milestone-based progress management, iterative improvement based on mentor feedback, and the consistent application of dashboard design best practices and data storytelling principles. This chapter provides a comprehensive and detailed account of every aspect of the project work undertaken during the internship.

5.1 Project Overview

The Food Trends: Understanding Customer Preferences in Food and Beverages project was conceived and developed to address a genuine and commercially significant business need in the Indian food and beverage industry — the need for a comprehensive, data-driven analytical tool that enables restaurant operators, food delivery platform managers, and business strategists to deeply understand their customers, products, operations, and markets in a way that supports smarter and more confidently evidence-based decision-making. In a business environment where data is generated at an unprecedented scale, the ability to transform raw transactional data into clear, actionable, and strategically relevant insights has become one of the most critical capabilities that any

food and beverage business can possess. The Indian food and beverage industry has undergone a profound transformation over the past decade, driven by the explosive growth of digital food delivery platforms such as Swiggy and Zomato, the rapid urbanization of the Indian population, the diversification of consumer tastes and preferences, and the intensification of competition among thousands of restaurants and food-tech companies operating on digital delivery platforms. Swiggy in particular has transformed food ordering from an occasional convenience into a daily habit for tens of millions of urban Indians, generating vast quantities of data about customer preferences, ordering behavior, satisfaction levels, and market dynamics in the process. It was within this rich and rapidly evolving industry context that the Food Trends project was designed and executed, with the fundamental premise that a well-designed Power BI dashboard built on genuine Swiggy restaurant data could provide stakeholders with a level of customer and market understanding that could directly translate into better menu decisions, smarter pricing strategies, more effective customer retention programs, and more efficient delivery operations.

The dataset used as the foundation for the project was sourced from Kaggle — the world’s most widely used open data science platform — where it was published as a publicly available dataset containing restaurant data scraped from the Swiggy platform across multiple major Indian cities including Bangalore and Hyderabad. This real-world data provenance was a crucial aspect of the project’s design, as it ensured that the insights generated by the dashboard reflected genuine patterns and dynamics of the actual Indian food delivery market rather than artificial or simulated data, while also introducing the real-world data quality challenges and analytical complexities that are inherent in any practical data analytics project. The primary objective of the project was to design and develop a professional-grade, multi-page interactive Power BI dashboard delivering a comprehensive 360-degree analytical view of the food and beverage landscape, covering all the key dimensions of business performance that matter most to industry stakeholders. This primary objective was broken down into seven specific analytical objectives, each corresponding to one of the seven report pages of the final dashboard — providing an executive performance summary, analyzing customer sentiment, uncovering food preferences and spending behavior, profiling customer segments, leveraging AI to identify satisfaction drivers, evaluating delivery operational performance, and delivering geographic market intelligence.

The dashboard was designed with a multi-audience perspective in mind, recognizing that the insights generated by the Swiggy dataset are relevant to a diverse range of stakeholders with different analytical needs. Senior executives needed a high-level market performance summary served by the Executive Overview page. Operational managers needed detailed delivery efficiency analysis served by the Operations and Delivery Performance page. Product managers needed cuisine preference and price sensitivity

insights served by the Product Insight page. Marketing professionals needed customer behavioral segmentation served by the Customer Segmentation page. Strategy teams needed AI-powered prescriptive analysis served by the Strategy and Innovation page. Regional managers needed geographic market intelligence served by the Geographic and Market Analysis page. The project scope encompassed the complete end-to-end technical development workflow — from data sourcing and download from Kaggle, through comprehensive data preparation in Power Query, data modeling, DAX measure development, visual design and dashboard layout, interactive feature implementation, and iterative refinement based on mentor feedback — culminating in a polished, professionally designed, and fully interactive Power BI dashboard ready for stakeholder presentation. The project was completed through structured milestones reviewed by Infosys Springboard program mentors, whose feedback was incorporated into successive development iterations, closely mirroring the agile project management approach used in professional data analytics teams and providing an authentic experience of how real-world analytical projects are planned, executed, and refined in a professional environment.

5.2 Dataset Description

The Food Trends project was built upon a comprehensive and richly structured real-world dataset capturing detailed restaurant-level and order-level information from the Indian food and beverage industry across multiple major cities. The dataset was sourced from Kaggle, where it was published as a publicly available dataset containing restaurant data scraped directly from the Swiggy platform — one of India's leading food delivery services connecting millions of customers with thousands of restaurants daily across hundreds of Indian cities. This real-world origin from the Swiggy platform ensured that the dataset reflected authentic restaurant listings, genuine customer ratings, real pricing information, and actual delivery time data, making it an ideal and credible foundation for a food trends analysis project. The raw dataset downloaded from Kaggle contained restaurant-level records from Swiggy's platform across multiple Indian cities including Bangalore and Hyderabad, with each record representing an individual restaurant listing and including a rich set of attributes capturing the restaurant's identity, cuisine offerings, pricing, customer ratings, delivery characteristics, and geographic location. The dataset was organized as a single unified fact table named Fact Orders within the Power BI data model, serving as the central analytical foundation of the entire dashboard.

The key variables in the dataset covered a wide and analytically rich range of dimensions. Restaurant Names provided the unique identity of each restaurant enabling restaurant-level analysis and ranking. Cuisines captured the type of cuisine offered

enabling preference and market share analysis. City identified the geographic location enabling city-level market comparisons. Price captured the average price point enabling pricing analysis and segmentation. Delivery Time recorded the average delivery duration in minutes enabling operational performance analysis. Average Ratings captured the mean customer rating enabling quality and satisfaction analysis. Sentiment provided a Positive, Neutral, or Negative classification of overall customer sentiment. Revenue captured total revenue generated enabling financial performance analysis. Order Frequency recorded how frequently customers ordered enabling engagement and loyalty analysis. Average Order Value captured the mean value of individual orders enabling spending behavior analysis.

ID	Area	City	Restaurant	Price	Avg ratings	Total ratings	Food type	Address
1	211	Koramangala	Tandoor Hut	300	4.4	100	Biryani	5TH Block
2	211	Koramangala	Tandoor Hut	300	4.4	100	Chinese	5TH Block
3	211	Koramangala	Tandoor Hut	300	4.4	100	North Indian	5TH Block
4	211	Koramangala	Tandoor Hut	300	4.4	100	South Indian	5TH Block
5	221	Koramangala	Tunday Kababi	300	4.1	100	Mughlai	5TH Block
6	221	Koramangala	Tunday Kababi	300	4.1	100	Lucknowi	5TH Block
7	246	Joghepalya	Kim Lee	650	4.4	100	Chinese	Double Road
8	248	Indiranagar	New Punjabi Hotel	250	3.9	500	North Indian	80 Feet Road
9	248	Indiranagar	New Punjabi Hotel	250	3.9	500	Punjabi	80 Feet Road
10	248	Indiranagar	New Punjabi Hotel	250	3.9	500	Tandoor	80 Feet Road
11	248	Indiranagar	New Punjabi Hotel	250	3.9	500	Chinese	80 Feet Road
12	249	Indiranagar	Nbh	350	4	50	Rajasthani	80 Feet Road
13	249	Indiranagar	Nbh	350	4	50	Gujarati	80 Feet Road
14	249	Indiranagar	Nbh	350	4	50	North Indian	80 Feet Road
15	249	Indiranagar	Nbh	350	4	50	Snacks	80 Feet Road
16	249	Indiranagar	Nbh	350	4	50	Desserts	80 Feet Road
17	249	Indiranagar	Nbh	350	4	50	Beverages	80 Feet Road
18	249	Indiranagar	Nbh	350	4	50	Thalis	80 Feet Road
19	249	Indiranagar	Nbh	350	4	50	Chaats	80 Feet Road
20	254	Indiranagar	Treat	800	4.5	100	Mughlai	100 Feet Road
21	254	Indiranagar	Treat	800	4.5	100	North Indian	100 Feet Road
22	258	Indiranagar	Chinita Real Mexican Food	1000	4.5	500	Mexican	Double Road
23	258	Indiranagar	Chinita Real Mexican Food	1000	4.5	500	Beverages	Double Road
24	258	Indiranagar	Chinita Real Mexican Food	1000	4.5	500	Salads	Double Road
25	263	Koramangala	Cupcake Noggins - Cakespastries And Desserts	150	4.3	100	Desserts	4TH Block
26	263	Koramangala	Cupcake Noggins - Cakespastries And Desserts	150	4.3	100	British	4TH Block
27	263	Koramangala	Cupcake Noggins - Cakespastries And Desserts	150	4.3	100	Bakery	4TH Block
28								

Fig 5.1: Dataset used for Food Trends analysis

The above figure illustrates the dataset preparation process in the Power Query Editor of Power BI. The dataset contains important attributes such as restaurant details, location information, pricing, ratings, food categories, and addresses used for the Food Trends Analysis project. Various data transformation operations were performed, including data type conversion, column renaming, data cleaning, and the creation of calculated fields such as the Price Segment category. These preprocessing steps improved data consistency, accuracy, and usability for analysis. The transformed dataset served as the foundation for data modeling, DAX calculations, dashboard development, and business intelligence reporting.

Additional derived variables engineered during data preparation included Price Segment classifying restaurants into Low, Mid-Range, and Premium pricing tiers. Rating Segment classifying restaurants into Poor, Average, Good, and Excellent quality tiers. Delivery Segment classifying restaurants as Fast, Standard, or Slow based on delivery speed. Customer Satisfaction Level providing an overall composite satisfaction assess-

ment. User Engagement Level classifying customers as Low, Medium, or High engagement. Fast Delivery Percentage capturing the proportion of orders delivered within the target time. Delivery Target indicating whether each restaurant consistently met its delivery time target. Top Performing City identifying the city with the highest composite performance score. This rich and multidimensional dataset provided the analytical raw material needed to address a wide range of business questions across all seven dashboard pages, from high-level market performance summaries to detailed operational efficiency analyses and granular customer segmentation profiles.

5.3 Data Preparation and Modeling

Before any report development could begin, the raw Swiggy dataset downloaded from Kaggle required a thorough and systematic process of data preparation and cleaning to transform it from its raw unprocessed state into a clean, consistent, complete, and analytically structured dataset suitable for building a professional Power BI dashboard. This work was performed entirely within the Power Query Editor in Power BI Desktop, where a carefully designed sequence of transformation steps was applied to address all identified quality issues and engineer the additional derived variables needed to support the full range of planned analyses. The data preparation process began with a comprehensive inspection of the raw dataset using Power Query's Column Quality, Column Distribution, and Column Profile features to systematically identify all quality issues requiring resolution. Duplicate restaurant records generated by the web scraping process were identified and removed using the Remove Duplicates function applied to the Restaurant Name and City combination, ensuring each restaurant appeared exactly once. Null and missing values across key columns including Price, Delivery Time, Average Ratings, and Revenue were addressed through column-specific strategies — numeric columns with sparse missing values were filled with column medians to preserve record counts, while records with missing values in critical fields were filtered out entirely to protect analytical integrity.

Incorrect data type assignments were corrected across multiple columns to ensure that numeric fields were treated as numbers and text fields were stored and processed correctly. Column names were reviewed and standardized to clear, professional, and consistently formatted naming conventions. Text columns were cleaned using Power Query's Trim and Clean functions to remove whitespace and invisible control characters that could cause inconsistencies in filtering and grouping operations. Conditional columns were engineered to create the Price Segment, Rating Segment, and Delivery Segment classifications that were essential for the segmentation analyses across multiple dashboard pages. Following the completion of all data preparation steps, the cleaned and enriched dataset was loaded into the Power BI data model structured around the

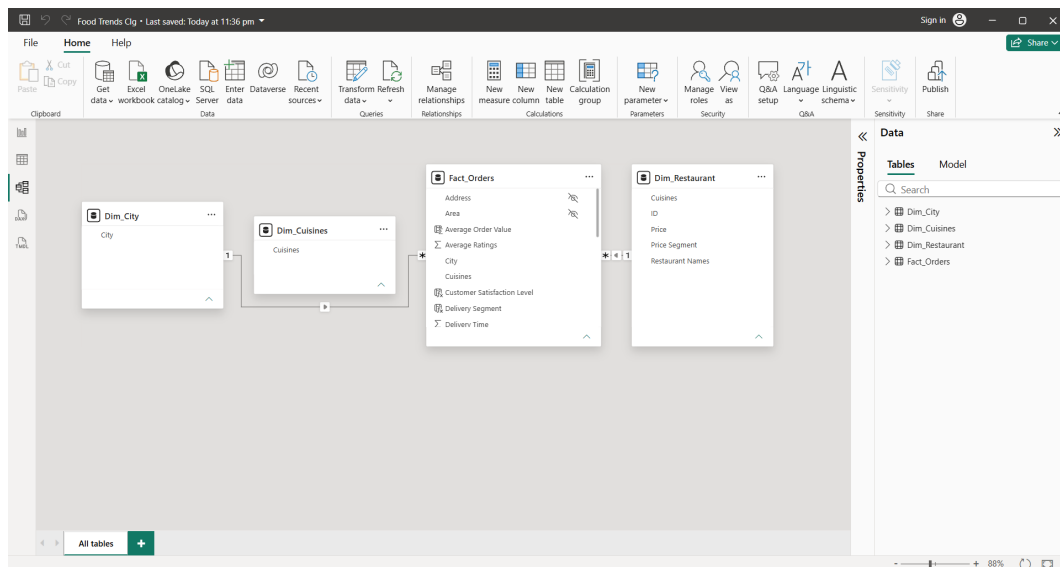


Fig 5.2: Star Schema Data Model

central Fact Orders table. A comprehensive suite of DAX measures was then developed to power all KPIs and analytical calculations across the seven report pages. Key measures included Total Orders using COUNTROWS, Total Restaurants using DISTINCTCOUNT on Restaurant Names, Total Cities using DISTINCTCOUNT on City, Average Rating using AVERAGEX, Average Price using AVERAGEX on the Price column, Average Delivery Time using AVERAGEX on the Delivery Time column, Fast Delivery Percentage using DIVIDE with CALCULATE and COUNTROWS filtered to Fast delivery records, Delivery Performance Score as a composite weighted measure, and Positive, Negative, and Neutral Restaurant Counts using CALCULATE with COUNTROWS filtered to each sentiment classification.

5.4 Executive Overview Page

The Executive Overview Page is the central dashboard of the Food Trends Analysis project and provides a complete summary of the overall business performance through advanced visualizations and interactive analytical reports. This page is designed to help business managers, stakeholders, and analysts monitor important organizational activities and understand overall operational performance in a simple and visually effective manner. It acts as the main control panel of the system where all major business indicators and trend reports are combined into a single interface for easier decision-making. The dashboard presents key business metrics such as Total Revenue, Total Orders, Total Customers, Customer Satisfaction Score, Average Delivery Time, Order Completion Rate, and Top-Selling Food Categories using KPI cards and summary visuals. These performance indicators help management evaluate business growth, identify operational strengths, and monitor customer engagement levels. The visual representation of data allows users to understand business conditions quickly without analyzing large volumes

of raw data manually.

The Executive Overview Page also provides detailed analysis regarding sales trends, customer purchasing patterns, product performance, and operational efficiency. Line charts and trend graphs are used to represent monthly and yearly sales growth, while category-wise reports help identify the food products generating the highest revenue and customer demand. Businesses can compare performance across different periods and evaluate whether business objectives are being achieved successfully. Interactive slicers and filters are integrated into the dashboard to improve report flexibility and user interaction. Users can analyze the data based on multiple parameters such as customer type, product category, delivery status, region, and time duration. These interactive features enable management to perform drill-down analysis and obtain detailed insights into specific business operations. The dashboard also supports strategic planning by providing insights into customer behavior, seasonal demand fluctuations, and market trends. Businesses can identify peak ordering periods, highly preferred cuisines, and customer engagement patterns through analytical reports. This information helps organizations improve marketing strategies, optimize inventory planning, and manage operational resources more effectively.

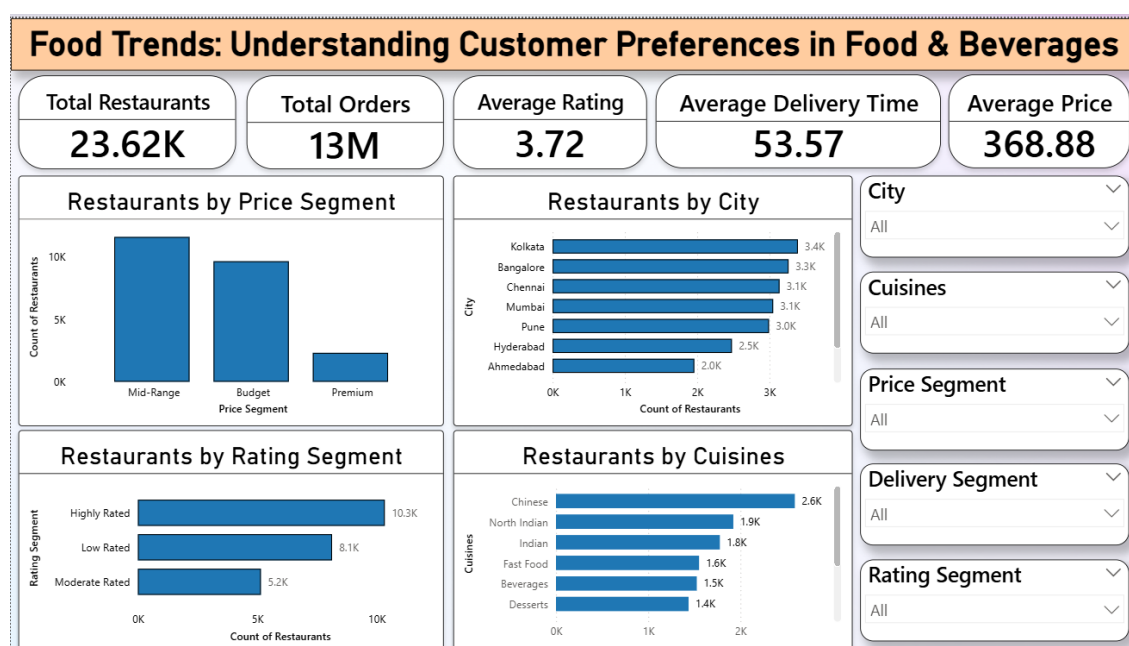


Fig 5.3: Executive Overview Dashboard

The above figure shows the Executive Overview page which serves as the command center of the Food Trends dashboard, displaying the most critical KPIs at a glance. The page presents eight key metrics — Total Restaurants, Total Orders, Total Cities, Average Rating, Average Price, Average Delivery Time, Top Performing City, and Dominant Satisfaction Level — through prominently formatted card visuals. Supporting charts show the distribution of restaurants by cuisine type, price segment breakdown, and

customer satisfaction distribution. Interactive City and Cuisine slicers allow users to filter the entire page dynamically. Navigation buttons enable direct access to all other report pages, making this page both the analytical entry point and central navigation hub of the dashboard.

Power BI visualization tools, DAX calculations, and Power Query transformations are extensively used to design the dashboard and ensure accurate analytical reporting. Data cleaning, transformation, and relationship modeling techniques are applied to improve data quality and report efficiency. The Executive Overview Page therefore acts as the foundation of the Food Trends Analysis system and supports data-driven business management and strategic decision-making. In addition, the dashboard improves communication between different business departments by presenting data in a standardized and visually understandable format. Sales teams, operations managers, and marketing departments can access the same dashboard to monitor performance and coordinate decision-making activities effectively. This reduces dependency on manual reporting processes and improves organizational productivity. The Executive Overview Page also helps identify business risks and operational challenges at an early stage. By monitoring KPIs regularly, management can detect declining customer satisfaction, delivery delays, or falling sales performance before they become major problems. Therefore, this dashboard not only supports business monitoring but also contributes to proactive business management and long-term organizational growth.

5.5 Sentiment Analysis Page

The Sentiment Analysis Page is developed to analyze customer feedback, reviews, and opinions collected from food ordering systems, online platforms, and customer surveys. The primary objective of this page is to understand customer emotions and satisfaction levels regarding food quality, pricing, delivery services, packaging, and overall customer experience. Sentiment analysis plays an important role in identifying customer expectations and improving business services according to customer feedback. The dashboard classifies customer opinions into three major sentiment categories: Positive, Neutral, and Negative sentiments. Positive sentiments represent customer satisfaction and appreciation for food quality, taste, freshness, hygiene, pricing, or fast delivery services. Negative sentiments indicate customer complaints related to delayed deliveries, poor packaging, bad food quality, inaccurate orders, or unsatisfactory customer service experiences. Neutral sentiments generally represent average customer opinions or mixed responses that do not strongly indicate satisfaction or dissatisfaction.

Various Power BI visualizations such as pie charts, bar graphs, sentiment trend charts, stacked column charts, and word cloud diagrams are used to represent customer emotions effectively. The word cloud visualization displays frequently used terms in

customer reviews, such as “delicious,” “fresh,” “excellent service,” “slow delivery,” “cold food,” and “poor packaging.” This helps management quickly identify common customer concerns and positive service aspects without reading thousands of individual reviews manually. Trend analysis charts are used to monitor changes in customer sentiment over different months, seasons, or promotional campaigns. Businesses can evaluate whether customer satisfaction is improving after introducing new products or operational improvements. Comparative sentiment analysis across different regions and customer groups also helps organizations understand geographic differences in customer preferences and expectations. The Sentiment Analysis Page supports customer relationship management by helping organizations improve service quality based on real customer experiences. Businesses can identify weak operational areas and implement corrective measures to enhance customer satisfaction and loyalty. Positive customer feedback can also be used for promotional campaigns and brand-building activities.

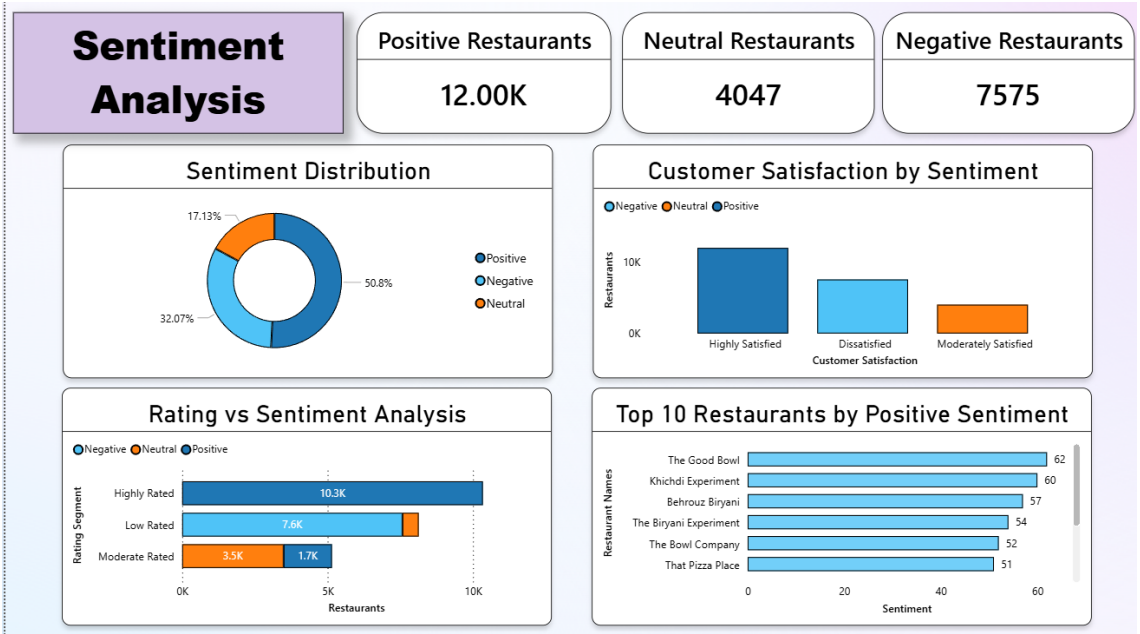


Fig 5.4: Sentiment Analysis Dashboard

The above figure shows the Sentiment Analysis page which provides a comprehensive analysis of customer perception and trust across the Swiggy restaurant dataset. Three KPI cards at the top display the count of Positive, Neutral, and Negative restaurants with color-coded formatting for instant interpretation. The Sentiment Distribution chart shows the proportional breakdown of restaurants across all three sentiment categories. The Rating vs. Sentiment Analysis visual explores the relationship between numerical ratings and sentiment classifications. The Customer Satisfaction by Sentiment visual reveals how satisfaction levels varied across sentiment categories. The Top 10 Restaurants by Positive Sentiment ranking identifies the highest-performing establishments from a customer trust perspective.

Advanced analytical techniques and DAX measures are used to calculate sentiment percentages, review frequencies, satisfaction scores, and customer engagement levels. Power Query is used to preprocess unstructured text data and remove irrelevant information before performing sentiment analysis. Data transformation processes improve analysis accuracy and reporting quality. The dashboard also helps management evaluate the impact of operational decisions on customer satisfaction. For example, businesses can monitor whether reducing delivery time leads to more positive customer feedback. Similarly, the impact of product pricing changes, promotional offers, or menu updates on customer sentiment can also be analyzed through the dashboard. Sentiment analysis additionally supports long-term business planning by helping organizations understand customer expectations and market trends. Businesses can identify emerging customer demands, improve product quality, and strengthen brand reputation through continuous feedback analysis. Overall, the Sentiment Analysis Page transforms raw customer opinions into valuable business intelligence that supports operational improvement, customer retention, and strategic growth.

5.6 Product Insight and Preference Analysis Page

The Product Insight and Preference Analysis Page focuses on analyzing customer food preferences, product performance, purchasing behavior, and revenue contribution of different food categories. This dashboard helps businesses understand which products are highly preferred by customers and which items contribute most significantly to business profitability. Product analysis is an important component of business intelligence because it supports effective product management, inventory planning, and marketing decision-making. The dashboard includes multiple visualizations such as category-wise sales charts, product ranking tables, revenue distribution graphs, trend analysis reports, and customer preference dashboards. These visual elements help businesses identify top-selling products, high-demand food categories, and seasonal purchasing patterns. Categories such as fast food, beverages, desserts, healthy meals, and traditional cuisines are analyzed separately to provide detailed insights into customer interests and market demand. Customer preference analysis is performed using historical order data, customer ratings, product reviews, and purchasing frequency patterns. Businesses can identify products that are consistently popular among customers and products that show declining demand over time. This analysis helps organizations understand changing food trends and customer behavior more accurately. The dashboard also provides insights regarding seasonal demand fluctuations and festival-related purchasing behavior. During festive seasons or special events, certain food categories may experience increased demand. Businesses can use this information to prepare inventory, optimize staffing, and improve operational planning to handle increased customer

orders effectively.

Low-performing products and declining categories are also identified through analytical reports. Businesses can evaluate possible reasons behind poor performance, such as pricing issues, quality concerns, or low customer awareness. Corrective actions such as product redesign, promotional offers, or menu improvements can then be implemented to improve product performance and increase customer satisfaction. Interactive slicers and drill-through capabilities allow users to analyze product performance based on customer segments, geographic regions, order types, and time periods. DAX calculations are used to measure product profitability, sales growth rate, average order value, and customer repeat purchase behavior. These analytical measures provide deeper business insights and support strategic decision-making.

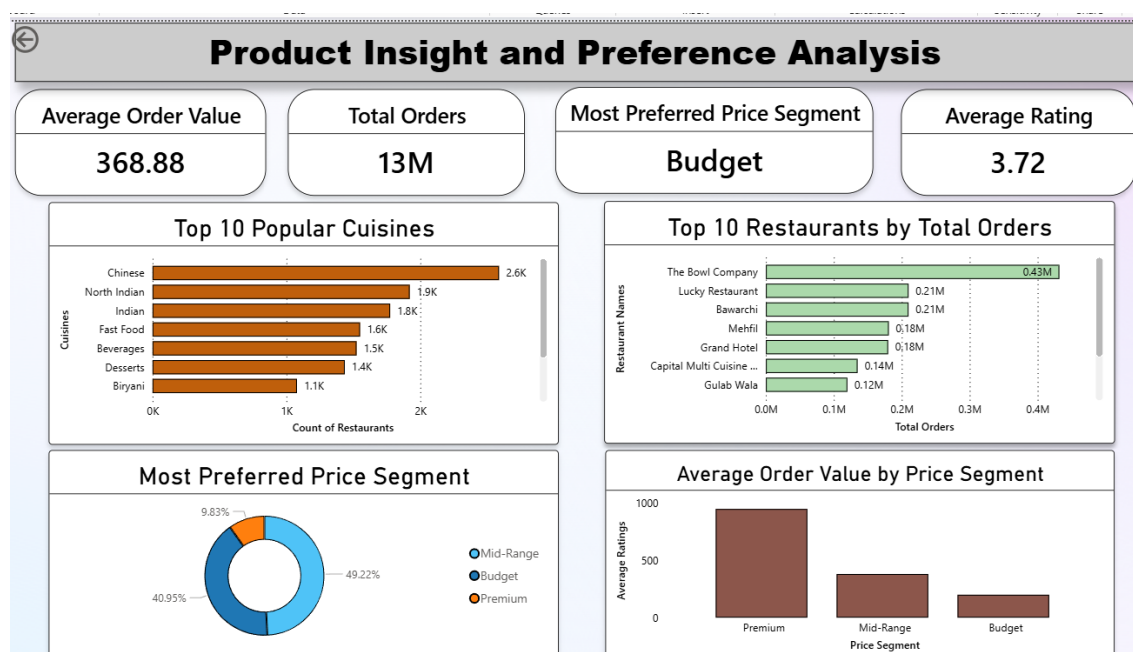


Fig 5.5: Product Insight and Preference Analysis Dashboard

The above figure shows the Product Insight and Preference Analysis page which uncovers customer ordering patterns, cuisine preferences, and spending behavior across the Swiggy platform. The Top 10 Popular Cuisines bar chart ranks the most frequently ordered cuisine types by total order volume. The Cuisine Market Share Analysis visual shows the proportional share of orders captured by each cuisine category. The Average Order Value by Price Segment chart reveals how customer spending varied across Low, Mid-Range, and Premium pricing tiers. The Restaurants by Price Segment and Restaurants by Rating Segment visuals show the supply-side distribution of restaurants across pricing and quality categories. The Most Preferred Price Segment card prominently displays the pricing tier attracting the highest order volume.

The Product Insight and Preference Analysis Page also supports innovation and new product development. Businesses can identify emerging food trends and customer

interests through demand analysis and customer feedback. This helps organizations introduce new menu items and personalized food recommendations according to market demand. Overall, this dashboard helps organizations improve product planning, maximize profitability, reduce wastage, and strengthen customer satisfaction by aligning business products with customer expectations and market trends.

5.7 Customer Segmentation Page

The Customer Segmentation Page is designed to classify customers into different groups based on their purchasing behavior, spending patterns, food preferences, order frequency, and demographic characteristics. Customer segmentation is one of the most important analytical techniques used in business intelligence because it helps organizations understand customer behavior more effectively and develop personalized business strategies. This dashboard enables businesses to identify valuable customer groups and improve customer relationship management through data-driven insights. The segmentation process divides customers into multiple categories such as Premium Customers, Loyal Customers, Frequent Buyers, Occasional Customers, New Customers, and High-Spending Customers. These customer groups are identified using factors such as total spending amount, number of orders placed, average order value, customer retention rate, and purchase frequency. Businesses can easily identify customers who contribute significantly to overall revenue and focus on retaining them through loyalty programs and personalized offers. The dashboard includes various visualizations such as customer distribution charts, demographic analysis graphs, cluster diagrams, heat maps, and spending pattern reports. These visual reports help management understand the behavior of different customer segments and analyze their contribution to business growth. Pie charts and bar graphs are used to represent the percentage contribution of each customer category to total sales and order volume.

Demographic analysis is also performed to study customer characteristics such as age group, gender, regional location, and preferred food categories. This analysis helps businesses understand customer interests and market demand in different geographic regions. For example, younger customers may prefer fast food and beverages, while health-conscious customers may show higher demand for healthy meal options and organic products. The dashboard also analyzes customer purchasing behavior and order frequency over different time periods. Businesses can identify which customers place orders regularly and which customers show declining engagement. This information helps organizations develop targeted marketing campaigns, promotional offers, and customer retention strategies to improve customer loyalty and increase sales performance. Interactive filters and slicers are integrated into the dashboard to enable detailed analysis based on region, customer type, order category, and time duration. Users can

compare customer segments across different locations and identify which markets generate the highest customer engagement. Drill-through reports provide deeper analysis of customer activities and help management understand customer buying behavior more accurately.

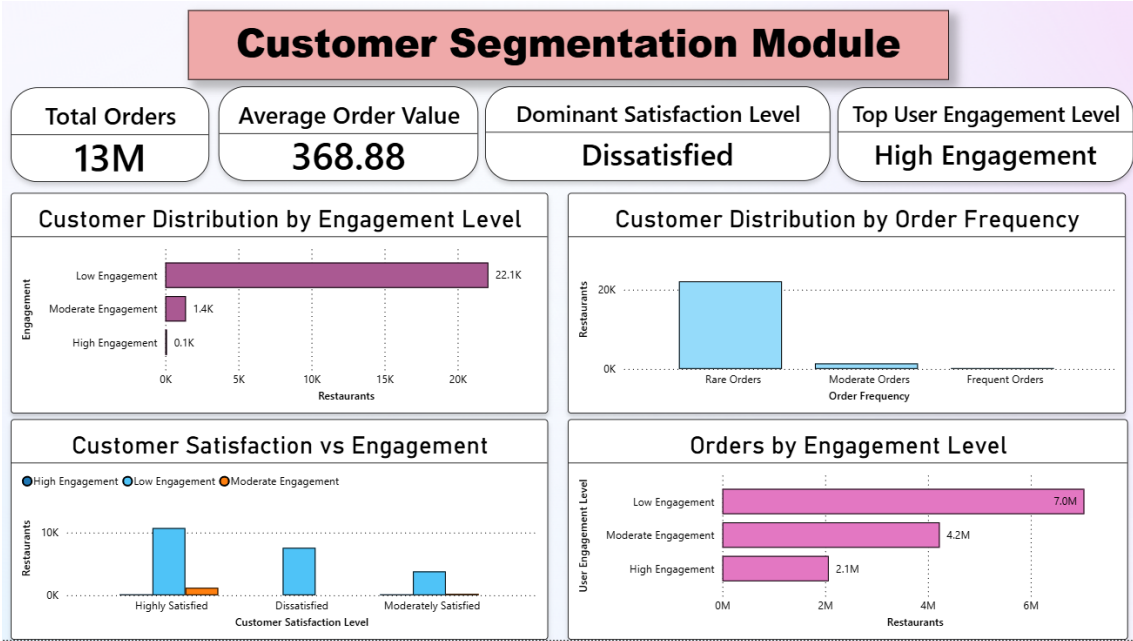


Fig 5.6: Customer Segmentation Dashboard

The above figure shows the Customer Segmentation page which provides a detailed behavioral profile of the Swiggy customer base across multiple engagement and satisfaction dimensions. The Customer Distribution by Engagement Level visual shows how customers were distributed across Low, Medium, and High engagement categories. The Customer Distribution by Order Frequency visual reveals the proportion of loyal habitual customers versus occasional ones. The Orders by Engagement Level chart demonstrates the disproportionately large contribution of highly engaged customers to overall order volumes. The Customer Satisfaction vs. Engagement visual confirms the positive relationship between engagement level and satisfaction outcomes. The Top User Engagement Level and Dominant Satisfaction Level card visuals provide prominent summary answers to the key customer profiling questions.

DAX calculations are used to measure important metrics such as Customer Lifetime Value (CLV), Average Purchase Frequency, Retention Rate, and Customer Profitability Index. These analytical measures provide valuable insights regarding customer contribution to business revenue and long-term profitability. Businesses can use these insights to prioritize valuable customer groups and improve customer satisfaction. The Customer Segmentation Page also supports personalized marketing and recommendation systems. Businesses can design customized offers and promotions for specific customer groups based on their purchasing history and preferences. Personalized mar-

keting improves customer engagement and increases the probability of repeat purchases and long-term customer retention. Overall, the Customer Segmentation Page helps organizations improve customer understanding, optimize marketing strategies, strengthen customer relationships, and maximize business profitability through advanced customer analytics and data-driven decision-making.

5.8 Strategy and Innovation Page

The Strategy and Innovation Page is developed to support long-term business growth, competitive planning, and innovation management within the food industry. This dashboard focuses on identifying emerging market trends, customer demands, business opportunities, and strategic improvement areas through advanced analytical reporting and trend analysis. The page helps businesses make future-oriented decisions and remain competitive in the rapidly changing food market environment. The dashboard analyzes customer purchasing behavior, product demand trends, market performance, and operational data to generate strategic insights for management. Businesses can identify which food categories are growing rapidly and which products are becoming less popular among customers. These insights help organizations modify business strategies according to changing market conditions and customer expectations. The Strategy and Innovation Page includes visualizations such as trend analysis charts, forecasting graphs, customer demand reports, product growth analysis, and strategic recommendation dashboards. These visual reports help management identify opportunities for product innovation, market expansion, and service improvement. Businesses can analyze historical data patterns and predict future business performance more accurately.

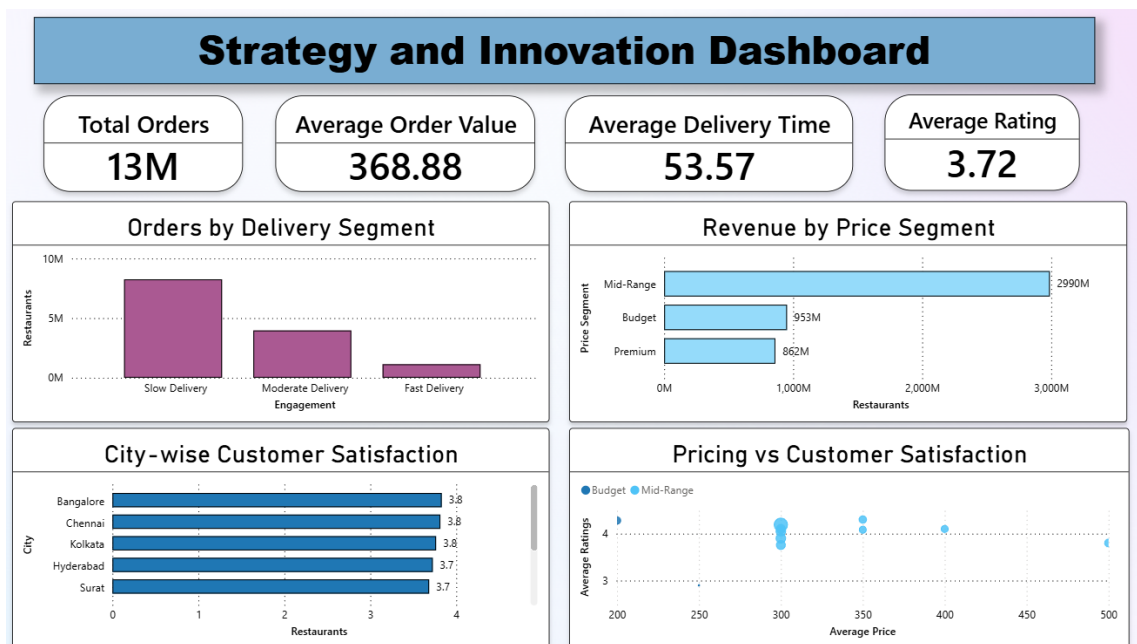


Fig 5.7: Strategy and Innovation Dashboard

The above figure shows the Strategy and Innovation page which leverages Power BI's built-in artificial intelligence capabilities to deliver prescriptive insights about the key drivers of customer satisfaction in the Swiggy dataset. The centerpiece of the page is the Key Influencers visual which uses machine learning algorithms to automatically identify and rank the variables most strongly predicting customer satisfaction outcomes. The left panel displays the ranked list of influencing factors including Price Segment, Delivery Segment, Cuisine type, and Engagement Level, while the right panel shows a detailed breakdown chart for the selected factor. The Top Segments view reveals the specific customer segments most strongly associated with high satisfaction outcomes.

One of the major objectives of this dashboard is to identify emerging food trends and innovation opportunities. The analysis may reveal increasing demand for healthy meals, organic products, low-calorie food options, or customized meal services. Businesses can use this information to introduce new menu items, improve existing products, and create innovative food experiences according to market demand. The dashboard also supports digital transformation and operational innovation by identifying areas where technology can improve customer experience and business efficiency. Businesses can evaluate opportunities such as AI-based recommendation systems, online ordering platforms, smart delivery tracking systems, personalized marketing automation, and customer engagement technologies. Forecasting techniques and predictive analytics are used to estimate future sales trends, customer demand patterns, and market growth opportunities. These predictive insights help management prepare for future challenges and optimize operational planning. Businesses can also estimate inventory requirements, staffing needs, and marketing budgets more effectively through demand forecasting.

Interactive filters and drill-through reports allow users to perform detailed analysis based on product category, customer segment, region, and time period. DAX calculations are used to calculate growth percentages, market contribution rates, customer demand indexes, and strategic performance indicators. These calculations support accurate business analysis and improve decision-making efficiency. The Strategy and Innovation Page also helps businesses evaluate competitor performance and market positioning. Organizations can identify gaps in the market and develop competitive strategies to attract more customers and increase market share. Strategic planning based on analytical insights improves business sustainability and long-term profitability. Overall, this dashboard supports innovation-driven business management by transforming analytical insights into actionable business strategies. It helps organizations improve operational efficiency, strengthen customer satisfaction, introduce innovative products, and achieve long-term growth through data-driven strategic planning.

5.9 Operations and Delivery Performance Page

The Operations and Delivery Performance Page is designed to monitor and evaluate operational efficiency, delivery performance, and service quality within the food business system. Efficient delivery operations and order management are essential for maintaining customer satisfaction and improving overall business performance. This dashboard helps organizations track operational activities, identify performance bottlenecks, and improve service efficiency through analytical reporting and real-time monitoring. The dashboard displays important operational KPIs such as Average Delivery Time, Total Orders Delivered, Delayed Deliveries, Order Completion Rate, Delivery Success Rate, Customer Complaint Count, and Delivery Partner Performance. These key performance indicators help management evaluate the effectiveness of operational processes and identify areas requiring improvement. Various Power BI visualizations such as line charts, performance gauges, heat maps, operational trend reports, and regional delivery analysis charts are used to represent operational data effectively. These visual reports allow management to monitor delivery performance across different cities, regions, and time periods. Businesses can identify which locations experience higher delivery delays and operational inefficiencies. The dashboard also analyzes peak order periods, workload distribution, and operational capacity utilization. Businesses can identify the busiest hours of the day and allocate staff, delivery partners, and resources more efficiently to manage high customer demand. This helps improve delivery speed, reduce waiting times, and enhance customer experience.

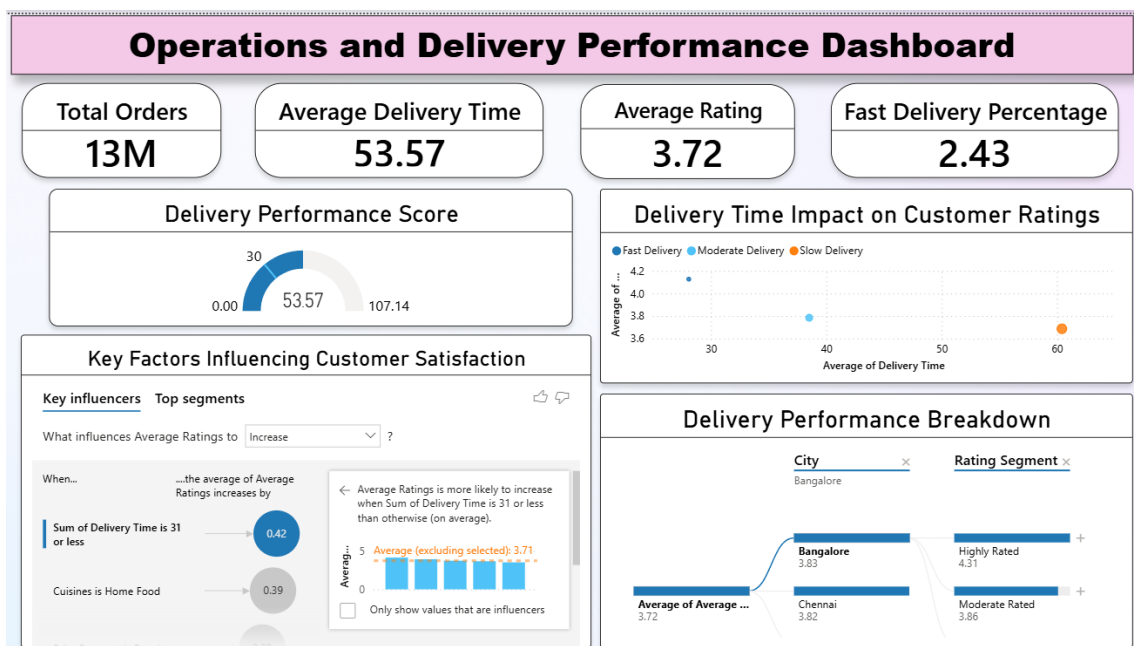


Fig 5.8: Operations and Delivery Performance Dashboard

The above figure shows the Operations and Delivery Performance page which provides a comprehensive analysis of delivery efficiency and its direct impact on customer

satisfaction across the Swiggy restaurant dataset. Three KPI cards at the top display Fast Delivery Percentage, Delivery Performance Score, and Average Delivery Time with conditional color formatting signaling performance against defined targets. The Delivery Time Impact on Customer Ratings visual quantifies the direct relationship between delivery speed and customer ratings, demonstrating how slower delivery was associated with measurably lower ratings. The Delivery Performance Breakdown visual decomposes delivery metrics across restaurants, cities, and cuisine types. The Orders by Delivery Segment chart shows the distribution of orders across Fast, Standard, and Slow delivery categories.

Operational performance analysis includes monitoring order processing time, preparation time, delivery route efficiency, and customer service response time. Businesses can identify operational bottlenecks that affect delivery quality and implement corrective measures to improve service efficiency. Reducing delivery delays directly contributes to improved customer satisfaction and positive customer sentiment. Interactive filters and slicers allow users to analyze delivery performance based on region, delivery partner, order type, and time duration. Drill-through reports provide detailed insights regarding operational challenges and help management evaluate individual delivery performance more accurately. DAX calculations are used to measure operational efficiency rates, average delivery delay, order processing performance, and customer service metrics. Power Query transformations are applied to clean and organize operational data before analysis. Accurate data processing ensures reliable reporting and analytical accuracy. The Operations and Delivery Performance Page also supports cost optimization and resource management. Businesses can analyze fuel usage, delivery expenses, staffing efficiency, and operational costs to improve profitability. Efficient operational management reduces unnecessary expenses and improves overall business productivity. Overall, this dashboard helps organizations improve operational efficiency, strengthen delivery performance, enhance customer satisfaction, and optimize resource utilization through data-driven operational analysis and business intelligence reporting.

5.10 Geographic and Market Analysis Page

The Geographic and Market Analysis Page is developed to analyze food trends, customer behavior, and business performance across different geographic regions and market segments. This dashboard helps organizations understand regional demand patterns, customer preferences, sales distribution, and market growth opportunities through location-based analytical reporting. Geographic analysis is an important component of business intelligence because customer preferences and market demand often vary significantly across different cities, states, and regions. The dashboard uses map visualizations, regional sales charts, heat maps, and geographic performance indicators

to represent location-based data effectively. Businesses can analyze sales performance across multiple regions and identify which locations generate the highest revenue and customer engagement. Geographic analysis helps organizations understand the relationship between location, customer demand, and business profitability.

The dashboard provides detailed insights regarding regional food preferences and customer purchasing behavior. Certain regions may show higher demand for fast food and beverages, while other areas may prefer healthy meals, desserts, or traditional cuisines. This analysis helps businesses customize menu offerings and marketing strategies according to regional customer interests and cultural preferences. The Geographic and Market Analysis Page also helps organizations identify high-performing markets and emerging business opportunities. Businesses can analyze areas with increasing customer demand and plan business expansion strategies accordingly. Regions with low sales performance can also be identified, allowing management to investigate possible reasons such as limited product availability, poor marketing, or operational inefficiencies. Interactive map visualizations are used to display customer density, order distribution, and regional sales contribution. Businesses can compare performance across different states, cities, or delivery zones and identify the most profitable markets. Regional comparisons support strategic planning and help organizations optimize resource allocation more effectively.

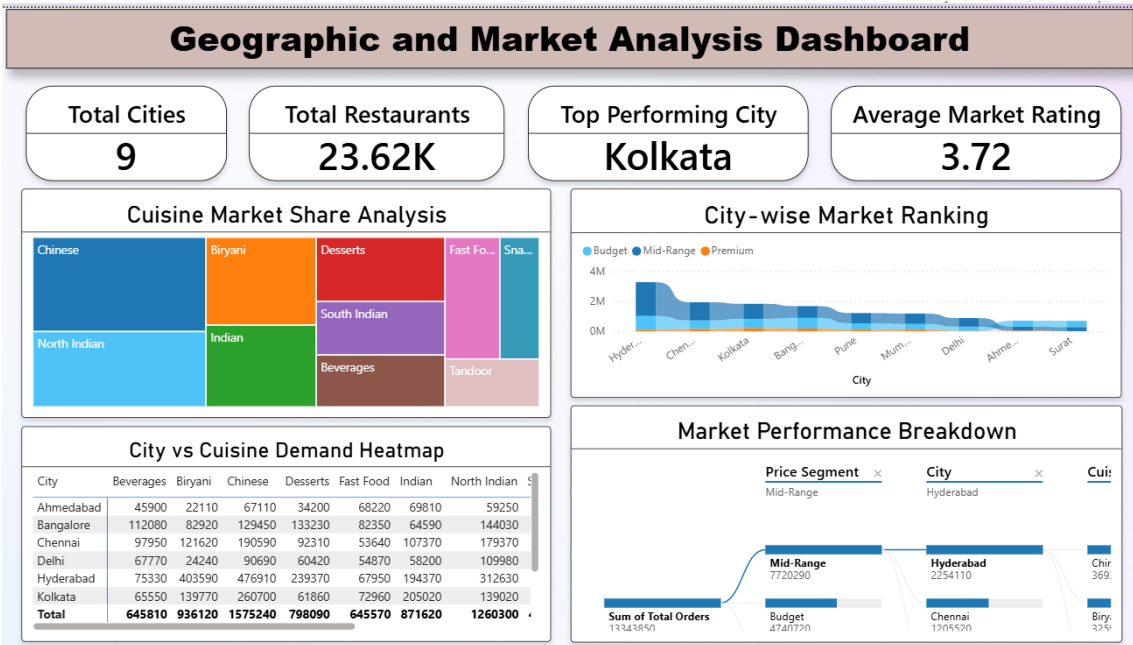


Fig 5.9: Geographic and Market Analysis Dashboard

The above figure shows the Geographic and Market Analysis page which brings a comprehensive location intelligence perspective to the Food Trends dashboard, enabling exploration of how restaurant performance and customer preferences varied across different Indian cities. The City vs. Cuisine Demand Heatmap displays a color-coded ma-

trix where darker colors represent higher cuisine demand concentrations across cities, enabling instant identification of the strongest geographic-cuisine demand patterns. The City-wise Customer Satisfaction visual enables direct comparison of satisfaction performance across geographic markets. The City-wise Market Ranking visual ranks all cities by their composite performance score combining average rating, order volume, revenue, and satisfaction. The Top Performing City, Average Market Rating, and Total Cities cards provide prominent summary reference points for the geographic analysis.

Market analysis reports are also included to evaluate customer demand trends, competitor influence, and market growth potential. Businesses can monitor changing customer preferences and emerging market trends through analytical reports. This information supports product innovation, targeted marketing campaigns, and customer-focused service improvements. The dashboard analyzes seasonal demand variations and location-based purchasing behavior. During festivals, holidays, or special events, customer ordering patterns may change significantly across different regions. Businesses can use these insights to prepare inventory, optimize delivery operations, and improve customer service during high-demand periods. Interactive slicers and filters allow users to perform detailed geographic analysis based on region, product category, customer segment, and time period. Drill-through reports provide deeper insights into regional performance and customer behavior. DAX calculations are used to measure regional sales growth, market contribution percentage, customer engagement rate, and geographic profitability indicators. Power BI map visualizations and data modeling techniques are extensively used to create accurate and visually understandable geographic reports. Power Query is used to clean and transform regional datasets before analysis to ensure data consistency and reporting accuracy. Overall, the Geographic and Market Analysis Page supports data-driven regional planning, market expansion strategies, customer-focused decision-making, and business growth by transforming location-based data into meaningful business insights and actionable recommendations. It enables organizations to identify profitable markets, understand regional customer preferences, and optimize business operations according to geographic demand patterns. The dashboard also helps management improve strategic planning, resource allocation, and targeted marketing activities for achieving sustainable business growth and competitive advantage in different market regions.

5.11 Key Insights and Findings

The Food Trends Analysis project generated several important insights and findings regarding customer behavior, product demand, operational performance, and market trends. These insights helped in understanding the overall business environment more effectively and supported strategic decision-making through data-driven analysis. One

of the major findings of the project was that fast food and beverage categories contributed the highest percentage of total sales and customer orders. These categories consistently showed strong customer demand across multiple regions and customer segments. Seasonal analysis also revealed that demand for certain food products increased significantly during festivals, weekends, and promotional campaigns. Customer sentiment analysis showed that positive customer feedback was strongly associated with fast delivery service, fresh food quality, accurate order processing, and attractive packaging. Customers were more satisfied when delivery times were reduced and service quality was maintained consistently. On the other hand, delayed deliveries and order inaccuracies were identified as major causes of negative customer feedback and customer dissatisfaction.

The project also revealed that Premium Customers and Loyal Customers contributed a major portion of total business revenue despite representing a smaller percentage of the overall customer base. These customers placed orders more frequently and showed higher average spending patterns compared to occasional customers. This finding highlighted the importance of customer retention strategies and loyalty programs for improving long-term profitability. Geographic analysis identified that urban areas generated significantly higher sales and customer engagement compared to rural regions. Certain regions showed strong demand for healthy meals and organic food products, indicating changing customer preferences toward health-conscious food options. Businesses can use this information to introduce region-specific products and customized marketing strategies. Operational analysis revealed that delivery efficiency directly affected customer satisfaction and repeat purchase behavior. Locations with faster delivery performance showed better customer ratings and higher customer retention rates. This insight emphasized the importance of optimizing operational processes and delivery management systems.

The project also identified that personalized marketing and customer segmentation techniques can significantly improve customer engagement and business performance. Customers responded positively to targeted promotional offers and personalized recommendations based on their purchasing history and preferences. Interactive Power BI dashboards and visualization techniques improved the efficiency of business reporting and helped convert large volumes of raw data into meaningful analytical insights. Management could monitor business performance more effectively through real-time dashboards and interactive reporting features. Another important finding was the growing popularity of healthy food categories and customized meal options among younger customer groups. This trend indicates changing market behavior and future opportunities for product innovation within the food industry. Overall, the project demonstrated that data analytics and business intelligence tools can significantly improve decision-making, operational efficiency, customer satisfaction, and strategic planning within the

food industry by transforming complex datasets into actionable business insights.

5.12 Challenges and Solutions

During the development and implementation of the Food Trends Analysis project, several technical, analytical, and operational challenges were encountered. These challenges affected data processing, dashboard development, analytical reporting, and system performance. Appropriate solutions and methodologies were implemented to overcome these issues and ensure successful project completion. One of the major challenges faced during the project was data inconsistency and poor data quality. The collected datasets contained duplicate records, missing values, incorrect formatting, and inconsistent naming conventions. Such issues affected analysis accuracy and dashboard reliability. To solve this problem, Power Query was used extensively for data cleaning and preprocessing. Duplicate records were removed, missing values were handled appropriately, and data formats were standardized to ensure consistency and accuracy. Another significant challenge was managing large volumes of data while maintaining dashboard performance and responsiveness. Large datasets increased report loading time and affected visualization performance. Efficient data modeling techniques, optimized relationships, and properly structured star schema architecture were implemented to improve dashboard efficiency and reduce system complexity.

Developing advanced DAX measures and analytical calculations was also a complex task. Several KPIs such as Customer Satisfaction Score, Delivery Efficiency Rate, Product Profitability, and Customer Lifetime Value required complex DAX formulas and conditional calculations. To overcome this challenge, different DAX functions such as CALCULATE, FILTER, SUMX, COUNTROWS, IF statements, and time intelligence functions were carefully implemented to generate accurate business metrics. Choosing suitable visualizations for representing complex business insights was another important challenge. Different types of business data required different visualization techniques for effective understanding and presentation. Appropriate visual elements such as KPI cards, line charts, heat maps, pie charts, stacked bar graphs, geographic maps, and trend reports were selected based on business requirements and analytical objectives. Performing sentiment analysis on customer reviews was also challenging because customer feedback data contained unstructured text, spelling variations, and mixed opinions. It was difficult to categorize reviews accurately into Positive, Neutral, and Negative sentiments. Text preprocessing techniques and sentiment classification methods were used to improve sentiment analysis accuracy and reporting quality. Maintaining dashboard interactivity while using multiple filters, slicers, and drill-through reports also affected performance optimization. Complex report structures sometimes reduced dashboard responsiveness.

CONCLUSION

The Food Trends Analysis project successfully demonstrated the importance of Business Intelligence and Data Analytics in improving decision-making, customer understanding, and operational efficiency in the food industry. By using Power BI, Power Query, DAX calculations, and data visualization techniques, the project converted raw data into meaningful insights and interactive dashboards for effective business analysis. The project provided detailed analysis of customer preferences, sentiment analysis, product performance, customer segmentation, delivery efficiency, and geographic market trends. Interactive dashboards and visual reports helped identify high-performing products, profitable customer segments, operational challenges, and emerging market opportunities. Features such as filters, slicers, KPI cards, charts, and maps improved reporting efficiency and supported detailed business analysis. The Sentiment Analysis dashboard helped understand customer satisfaction levels regarding food quality, pricing, packaging, and delivery services. Similarly, Product Insight and Customer Segmentation dashboards supported personalized marketing strategies and customer-focused business planning. Geographic and Market Analysis further helped businesses understand regional demand patterns and optimize market expansion strategies.

Several challenges such as data cleaning, handling large datasets, dashboard optimization, and visualization selection were successfully resolved using appropriate analytical techniques and Power BI tools. These challenges improved practical understanding of Business Intelligence implementation and dashboard development. Overall, the Food Trends Analysis project proved that data-driven analysis and visualization techniques can improve business performance, customer satisfaction, operational management, and strategic planning in the food industry. The project also enhanced practical knowledge of Power BI and Business Intelligence tools and serves as a strong foundation for future advanced analytics systems. Additionally, the project highlighted the importance of using modern Business Intelligence tools for transforming complex datasets into simple and visually understandable reports. The implementation of interactive dashboards improved the efficiency of monitoring business performance and enabled better communication of insights to management. The project also developed practical skills in data modeling, report generation, analytical thinking, and dashboard design, which are essential for real-world business analytics and decision-making processes.

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